

Semiconductor Design Infrastructure Handbook

Deployment, Configuration, and Administration
of Cadence-based RHEL workstations.

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Abstract

This manual presents a beginner-oriented but infrastructure-conscious guide to building and using a Cadence-based semiconductor design environment on Red Hat Enterprise Linux. It begins by explaining why modern Electronic Design Automation workflows depend on enterprise Linux platforms, why RHEL and RHEL-compatible systems are commonly used in academic and industrial VLSI environments, and how a laboratory workstation must be planned so that the operating system, user identity, licensing, storage, and tool environment remain consistent across machines.

The first part of the document focuses on environment preparation. It covers the practical decisions required before installing RHEL, including hardware expectations, virtualization versus native dual booting, installation media preparation, UEFI and Secure Boot considerations, disk partitioning strategy, and post-installation workstation identity. It then introduces the configuration work required for Cadence usage, including directory organization, environment variables, C-shell setup, license configuration, installation verification, and tool launch procedures. This establishes the workstation as a usable and reproducible EDA platform rather than as a generic Linux desktop.

After the system environment is established, the manual transitions into the digital design flow itself. It introduces Verilog, combinational and sequential RTL design, testbench construction, simulation, waveform analysis, and coverage analysis using Cadence tools such as Xcelium and IMC. The learning path is centered around small laboratory exercises, including adders and counters, so that each concept is connected to a concrete design task rather than presented only as theory.

The later parts of the document extend the same design from RTL toward implementation. The manual outlines logic synthesis using Genus, timing analysis, logical equivalence checking with Conformal, and the major stages of physical design using Innovus, including floorplanning, power planning, placement, clock tree synthesis, routing, and physical verification. It concludes with GDSII generation, tapeout context, the relationship between silicon and PCB-level integration, and a complete review of the RTL-to-GDSII flow.

Overall, the intent of this manual is to give new users a coherent view of both sides of an EDA laboratory: the infrastructure required to run professional design tools reliably, and the step-by-step design flow used to take a simple digital circuit from Verilog source code through simulation, coverage, synthesis, verification, physical implementation, and final layout data generation.

Preface

The ability to design integrated circuits has become one of the defining technological capabilities of the modern world. From communication systems and transportation infrastructure to medical devices, defense electronics, artificial intelligence accelerators, and high-performance computing platforms, nearly every advanced engineering discipline today relies upon semiconductor technology. Yet the process of designing semiconductor devices is not enabled by design software alone. Behind every successful VLSI laboratory exists a carefully constructed computing infrastructure capable of supporting Electronic Design Automation (EDA) workflows reliably, consistently, and at scale.

The creation of such an environment is rarely visible to students. When a laboratory workstation is switched on and a design tool launches successfully, years of operating system development, network engineering, software integration, licensing infrastructure, and administrative planning remain hidden beneath the surface. However, establishing that environment from scratch is often a complex undertaking involving far more than the installation of a single software package.

This document originated from the practical challenges encountered during the establishment of the semiconductor design infrastructure used within the Department of Electronics and Communication Engineering at Gati Shakti Vishwavidyalaya. The work described throughout this manual was carried out as part of a broader effort to deploy a standardized Linux-based CAD environment capable of supporting Cadence design tools under the Chips-to-Startup (C2S) Programme.

The deployment process involved the installation and configuration of Red Hat Enterprise Linux workstations, preparation of user environments, integration of licensing services, workstation provisioning, software deployment, shell configuration, network validation, and the establishment of a repeatable administrative workflow. While many of these tasks appear straightforward when viewed individually, their interaction within a laboratory environment introduces a level of complexity that is often underappreciated until deployment begins.

A significant portion of this work was carried out by the author under the guidance of Dr. Prasiddh Trivedi, Assistant Professor, Department of Electrical Engineering, whose support, technical direction, and institutional coordination made the deployment possible. Throughout the setup process, numerous infrastructure decisions had to be evaluated and justified, ranging from operating system selection and workstation architecture to user naming conventions, deployment methodologies, storage allocation strategies, and license-compliance requirements. The resulting environment was not assembled through trial and error alone, but through a deliberate attempt to create a platform that could be maintained, reproduced, and expanded as the laboratory evolved.

One of the recurring observations during the deployment was the scarcity of documentation describing the complete setup process from an institutional perspective. Vendor manuals typically explain how to install a particular software package, while operating system documentation explains how to configure a particular service. What is often missing is a unified narrative describing how these components fit together within a real laboratory environment. Future administrators are therefore frequently forced to reconstruct deployment decisions by examining existing machines, consulting fragmented notes, or relying upon institutional memory.

This manual was written to address that problem.

Its purpose is not merely to record commands that were executed during installation, but to capture the reasoning behind those decisions. Wherever possible, procedures are accompanied by explanations describing why a particular configuration was chosen, what assumptions were made, and what consequences may arise if those assumptions change. The objective is to transform administrative knowledge from an informal collection of personal experience into a structured and reproducible reference.

The need for such documentation becomes especially important in academic environments. Students graduate, research projects conclude, hardware is replaced, operating systems reach end-of-life, and administrative responsibilities change hands. Without documentation, knowledge accumulated during deployment can disappear remarkably quickly. A workstation that functions perfectly today may become difficult to rebuild several years from now if the decisions that produced it were never recorded. By documenting both procedures and rationale, the laboratory reduces its dependence on individual administrators and preserves

institutional knowledge for future use.

At the time of writing, many of the systems described in this document are already being deployed directly rather than reconstructed from documentation. Consequently, it is entirely possible that no immediate reader will ever need to follow every procedure presented here from beginning to end. In fact, the most successful outcome may be that future administrators rarely need this manual at all because the infrastructure continues to operate smoothly. Nevertheless, the value of technical documentation should not be measured solely by how often it is consulted. Its true purpose is to exist when something eventually changes: when a workstation fails, when a laboratory expands, when software must be reinstalled, or when a future administrator asks a simple but important question—"How was this system originally set up?"

This handbook therefore serves both as an operational reference and as a historical record of the deployment effort undertaken to establish the semiconductor CAD environment at Gati Shakti Vishwavidyalaya. It documents not only the final configuration of the systems, but also the engineering considerations, constraints, and administrative decisions that shaped their design.

The chapters that follow describe the deployment of the workstation environment, operating system preparation, configuration standards, and supporting infrastructure required for Cadence-based semiconductor design workflows. Together, they represent an attempt to ensure that the knowledge acquired during this deployment remains accessible, reproducible, and useful long after the initial installation work has been completed.

Contents

Abstract	i
Preface	ii

I SYSTEM SETUP 1

1. Foundations of the EDA Environment	3
1.1 Introduction to EDA	3
1.2 Why Enterprise Linux Anchors Modern EDA?	3
1.3 Red Hat Enterprise Linux as Industrial Infrastructure	4
1.4 Design Philosophy of the EDA Environment	4
1.5 Overview of Laboratory Infrastructure	5
1.6 Workstation Hardware Requirements and Deployment Constraints	6
1.6.1 Storage Requirements	6
1.6.2 Memory Requirements	7
1.6.3 Deployment Methodology: Native Installation versus Virtualization	7
1.7 Summary	8
2. RHEL Installation and Setup	9
2.1 Installation Planning and Deployment Strategy	9
2.2 Hardware Requirements and Workstation Specifications	10
2.3 Obtaining the RHEL Installation Media	11
2.4 BIOS, UEFI and Secure Boot Configuration	11
2.5 Virtualized RHEL Installation	12
2.5.1 Preparing the Virtual Machine	12
2.6 Native Dual Booting RHEL 8.7	16
2.6.1 Disk Partitioning Strategy	16
2.6.2 Creating the Installation Media	18
2.6.3 Installation Destination Stage	19
2.7 Chapter Summary	20
3. Post Install Configurations	21
3.1 Username/Hostname Configuration	21
3.2 Installing and Configuring Cadence	23
3.2.1 Cadence Directory Structure	23
3.2.2 Environment Variables	24
3.2.3 C-Shell Configuration	24
3.2.4 License Configuration	25

3.2.5 Verifying Installation	26
3.2.6 Launching Cadence Tools	27

II RTL DESIGN FUNDAMENTALS 29

4. Introduction to Verilog	31
4.1 Why Hardware Description Languages Exist	31
4.2 Verilog Program Structure	31
4.3 Modules	32
4.4 Ports	32
4.5 Nets and Registers	33
4.6 Numbers and Constants	33
4.7 Operators	34
4.8 Laboratory Exercise – Half Adder	34
4.8.1 Objective	34
4.8.2 Theory	35
4.8.3 Truth Table	35
4.8.4 Verilog Implementation	35
4.8.5 Testbench	36
4.8.6 Simulation Procedure	36
4.8.7 Expected Results	37
4.8.8 Conclusion	37
5. Combinational Logic Design.	39
5.1 Continuous Assignments	39
5.2 Dataflow Modeling	40
5.3 Conditional Operators	41
5.4 Multiplexers	43
5.4.1 2:1 Multiplexer	43
5.4.2 4:1 Multiplexer	44
5.5 Full Adder Design	45
5.6 Laboratory Exercise – 4-Bit Ripple Carry Adder	46
5.6.1 Objective	46
5.6.2 Theory	46
5.6.3 Ripple Carry Adder Architecture	47
5.6.4 Full Adder Module	47
5.6.5 4-Bit Ripple Carry Adder Implementation	48
5.6.6 Testbench Development	49
5.6.7 Simulation Procedure	49
5.6.8 Expected Results	50
5.6.9 Conclusion	50
6. Sequential Logic Design.	51
6.1 Why Memory is Required	51
6.2 Clock Signals	52
6.3 D Flip-Flops	53
6.4 Registers	54
6.5 Counters	55

6.6	Laboratory Exercise – 4-Bit Synchronous Counter	57
6.6.1	Objective	57
6.6.2	Theory	57
6.6.3	Counter Architecture	57
6.6.4	Verilog Implementation	58
6.6.5	Testbench Development	58
6.6.6	Simulation Procedure	59
6.6.7	Expected Results	59
6.6.8	Conclusion	60

III VERIFICATION 61

7.	Verilog Testbenches	63
7.1	What is Verification?	63
7.2	Testbench Architecture	63
7.3	Initial Blocks	63
7.4	Always Blocks	63
7.5	Clock Generation	63
7.6	Stimulus Generation	63
7.7	Self-Checking Testbenches	63
7.8	Laboratory Exercise	63
8.	Simulation and Waveform Analysis	65
8.1	Compilation	65
8.2	Elaboration	65
8.3	Simulation	65
8.4	Understanding Waveforms	65
8.5	Common RTL Bugs	65
8.6	Debugging Methodology	65
8.7	Laboratory Exercise	65
9.	Coverage Analysis using IMC	67
9.1	Why Simulation Alone is Not Enough	67
9.2	Code Coverage	67
9.3	Branch Coverage	67
9.4	Toggle Coverage	67
9.5	Functional Coverage	67
9.6	Using IMC	67
9.7	Coverage Reports	67
9.8	Coverage Closure	67
9.9	Laboratory Exercise	67

IV LOGIC SYNTHESIS 69

10.	Introduction to Synthesis	71
10.1	RTL vs Hardware	71
10.2	Standard Cell Libraries	71
10.3	Liberty Files	71
10.4	Timing Constraints	71

10.5 The Synthesis Flow	71
10.6 Reading Reports	71
11. Logic Synthesis using Genus	73
11.1 Preparing RTL	73
11.2 Writing Constraints	73
11.3 Running Genus	73
11.4 Generated Outputs	73
11.5 Laboratory Exercise	73
12. Timing Analysis	75
12.1 Propagation Delay	75
12.2 Setup Time	75
12.3 Hold Time	75
12.4 Clock Frequency	75
12.5 Critical Paths	75
12.6 Slack	75
12.7 Laboratory Exercise	75
13. Logical Equivalence Checking.	77
13.1 Why Equivalence Checking Matters	77
13.2 Golden vs Revised Designs	77
13.3 Conformal Workflow	77
13.4 Laboratory Exercise	77
V PHYSICAL DESIGN	79
14. Introduction to Physical Design	81
14.1 What Happens After Synthesis?	81
14.2 Netlists and Standard Cells	81
14.3 Die, Core and IO Regions	81
14.4 Physical Design Flow Overview	81
15. Floorplanning	83
15.1 Defining the Core Area	83
15.2 IO Placement	83
15.3 Macro Placement	83
15.4 Utilization	83
15.5 Laboratory Exercise	83
16. Power Planning	85
16.1 Why Chips Need Power Networks	85
16.2 Power Rings	85
16.3 Power Stripes	85
16.4 IR Drop Concepts	85
16.5 Ground Networks	85
16.6 Laboratory Exercise	85
17. Placement	87
17.1 Standard Cell Placement	87

17.2 Congestion	87
17.3 Density Optimization	87
17.4 Laboratory Exercise	87
18. Clock Tree Synthesis	89
18.1 Clock Distribution	89
18.2 Clock Buffers	89
18.3 Clock Skew	89
18.4 Clock Latency	89
18.5 Laboratory Exercise	89
19. Routing	91
19.1 Global Routing	91
19.2 Detailed Routing	91
19.3 Routing Resources	91
19.4 Signal Integrity	91
19.5 Laboratory Exercise	91
20. Physical Verification	93
20.1 Design Rule Checking (DRC)	93
20.2 Layout Versus Schematic (LVS)	93
20.3 Common Physical Violations	93
20.4 Laboratory Exercise	93
VI TAPEOUT AND BEYOND	95
21. GDSII Generation	97
21.1 What is GDSII?	97
21.2 Manufacturing Data	97
21.3 Tapeout Process	97
22. From Silicon to PCB	99
22.1 Packaging	99
22.2 Pin Assignment	99
22.3 Chip Integration	99
22.4 PCB Design Workflow	99
22.5 Why PCB Design is Different from ASIC Design	99
23. Complete Design Flow Review	101
23.1 Revisiting the Counter	101
23.2 RTL → Simulation → Coverage	101
23.3 Synthesis → Timing → Verification	101
23.4 Floorplanning → CTS → Routing	101
23.5 Tapeout	101
23.6 Final Flow Summary	101
A. c2ssetup Script for Automating User/Hostname Config.	103
B. Cshrc configuration code	107

Part I

SYSTEM SETUP

Operating System Deployment and RHEL Workstation Initialization

01

Foundations of the EDA Environment

1.1 Introduction to EDA

Electronic Design Automation (EDA) refers to the collection of software tools, computational frameworks, and engineering workflows used in the design, simulation, verification, and physical implementation of integrated circuits and semiconductor systems. Modern semiconductor devices contain millions to billions of transistors operating at extremely small geometries, making manual design methodologies computationally impractical and operationally impossible. EDA tools therefore serve as the foundational infrastructure enabling engineers to transform conceptual circuit designs into manufacturable silicon devices.

Contemporary EDA environments encompass a wide range of specialized applications including schematic capture systems, Hardware Description Language (HDL) development environments, circuit simulators, synthesis tools, physical layout editors, parasitic extraction frameworks, timing analysis engines, and design-rule verification systems. These tools collectively support the complete semiconductor development lifecycle, from behavioral modeling and functional verification to physical layout generation and fabrication validation.

The increasing complexity of Very Large Scale Integration (VLSI) systems has made EDA indispensable within both academic and industrial semiconductor design workflows. Beyond improving productivity, EDA enables precision, scalability, repeatability, and verification accuracy that would otherwise be unattainable through conventional design methodologies. As a result, modern semiconductor engineering is fundamentally dependent upon robust EDA infrastructure and the computational environments that support it.

1.2 Why Enterprise Linux Anchors Modern EDA?

Modern EDA toolchains impose demanding requirements on the underlying operating system environment, including stability, deterministic behavior, scripting flexibility, network interoperability, multi-user access control, and long-term software compatibility. Consequently, semiconductor design organizations and research laboratories overwhelmingly deploy EDA infrastructure on Linux-based operating systems rather than general-purpose desktop platforms.

Linux provides a highly modular and administratively transparent environment capable of supporting complex engineering workflows involving distributed compute resources, centralized licensing systems, shared storage infrastructure, remote administration, and automated deployment methodologies. Native shell scripting capabilities, package management systems, process control utilities, and network tooling further enable scalable workstation administration and reproducible environment configuration across large laboratory deployments.

Among available Linux distributions, enterprise-grade platforms such as **Red Hat Enterprise Linux (RHEL)** and its derivatives have become the de facto standard within semiconductor design environments. Their widespread adoption is driven by long-term support lifecycles, predictable release stability, extensive vendor certification, and broad compatibility with commercial EDA applications. Most major semiconductor

tool vendors officially validate and support their software primarily on enterprise Linux environments, making RHEL-based systems the preferred foundation for reliable and maintainable CAD infrastructure deployment.

Accordingly, the workstation architecture described throughout this manual is built upon a standardized RHEL-based environment designed to provide operational consistency, scalability, and long-term maintainability for semiconductor design laboratories and VLSI research workflows.

1.3 Red Hat Enterprise Linux as Industrial Infrastructure

The decision to standardize a semiconductor design environment on a specific Linux distribution is not a matter of administrative preference. It is an infrastructure decision with direct consequences for EDA tool compatibility, operational stability, vendor support, and long-term maintainability. Within professional semiconductor environments, Red Hat Enterprise Linux has become a preferred foundation because it provides the stability and compatibility characteristics required by commercial EDA workflows.

Commercial EDA tools are typically distributed as precompiled binary applications that depend on stable system libraries, predictable runtime behavior, and well-defined operating system interfaces. RHEL maintains Application Binary Interface and Application Programming Interface consistency within a major release, allowing validated tool installations to remain reliable across routine system updates. This consistency is especially important for tools that depend on components such as `glibc`, `libX11`, Motif libraries, OpenSSL, shell utilities, and kernel-level behavior.

RHEL also provides long-term lifecycle stability that aligns well with semiconductor design workflows. A laboratory or production design environment may depend on a specific operating system and EDA tool combination for several years. Frequent distribution-level changes can introduce dependency drift and invalidate previously working configurations. By contrast, RHEL's enterprise release model prioritizes predictable maintenance, security updates, and controlled package evolution over rapid upstream change.

Vendor certification is another important factor. Major EDA vendors commonly certify their tools against specific RHEL major versions and closely compatible derivatives. Operating within a certified platform reduces support risk, simplifies troubleshooting, and provides a defensible baseline when vendor escalation is required. While unsupported Linux distributions may sometimes run the same tools, they introduce additional uncertainty into installation, execution, and support workflows.

The RHEL ecosystem also supports maintainable administration practices through standard tools such as `dnf`, `systemd`, `lvm`, `sssd`, and `subscription-manager`. These tools provide a well-documented administrative surface for package management, service control, storage configuration, authentication integration, and system maintenance. In institutional environments where systems must be rebuilt, audited, extended, or handed over between administrators, this consistency is a practical advantage.

Throughout the remainder of this manual, operating system procedures, configuration examples, package names, service management commands, and administrative workflows are specified for a RHEL 9-compatible environment. Unless otherwise noted, the procedures also apply to Rocky Linux 9 and AlmaLinux 9 as binary-compatible RHEL derivatives. References to *RHEL* in the operational context of this document should therefore be understood to include these derivatives unless a Red Hat subscription-specific feature is explicitly required.

1.4 Design Philosophy of the EDA Environment

The semiconductor design infrastructure described in this manual is not assembled opportunistically. It is engineered according to a defined set of operational principles that govern every decision from initial workstation deployment through long-term maintenance. Understanding these principles is necessary context for the procedures that follow, because many configuration choices that might otherwise appear arbitrary are direct expressions of the underlying philosophy.

The environment is built around **standardization**. All workstations in the laboratory are deployed from a common baseline: identical operating system version, identical package selection, identical directory structures, and identical tool installation paths. Deviation from this baseline is treated as an exception requiring justification, not a routine outcome of per-machine administration. Standardization is the prerequisite for everything else.

From standardization follows **reproducibility**. A workstation rebuilt from documented procedures should produce an environment functionally identical to the one it replaces. This property matters both operationally—when a workstation fails and must be restored—and institutionally, as the laboratory’s administrative knowledge must remain embedded in its documentation rather than in the accumulated state of individual machines.

The deployment is designed to **scale**. Procedures written for a single workstation apply without modification to the full laboratory fleet. Configuration that must be individually customized per machine is a design defect. Where centralized mechanisms exist—for user authentication, license server connectivity, shared storage, or environment initialization—they are used in preference to local equivalents.

Centralized administration follows from this. User accounts, access permissions, and shell environments are managed at the infrastructure level rather than configured locally on each workstation. This reduces the administrative surface area, eliminates inconsistency across machines, and ensures that changes propagate uniformly rather than requiring repeated per-host intervention.

Underlying all of these principles is the explicit goal of **eliminating configuration drift**. In long-running laboratory environments, workstations tend to diverge from one another over time as ad hoc fixes, local experiments, and undocumented modifications accumulate. The procedures in this manual are written to resist this tendency—through documented baselines, version-controlled configuration, and periodic validation practices that detect and correct deviation before it compounds.

The result is an infrastructure in which any workstation can be administered, diagnosed, or rebuilt by any administrator with access to this document, without dependence on institutional memory or machine-specific knowledge.

1.5 Overview of Laboratory Infrastructure

The laboratory environment should be understood as more than a collection of independent desktop computers. In a semiconductor design lab, each workstation is one part of a larger operating environment that includes user accounts, network identity, license access, shared design data, tool installation paths, and common shell configuration. A student may experience the system as “logging into Linux and opening Cadence”, but behind that simple action several infrastructure components must work together reliably.

At the center of the setup are the **RHEL workstations**. These are the machines on which students and users log in, write Verilog, launch simulations, open waveforms, run synthesis, and eventually interact with physical design tools. Each workstation may be installed natively or may initially run RHEL inside a virtual machine, but its role is the same: it acts as the user’s EDA compute endpoint. The workstation must therefore have a predictable hostname, a correctly named user account, a stable operating-system baseline, and access to the required Cadence environment.

Surrounding the workstations are the shared services that make the lab behave like a coordinated engineering environment rather than a set of isolated PCs. The **license server** controls access to Cadence tools; the workstation requests permission from this server whenever a licensed tool is launched. Shared storage, when available, provides a common location for project files, PDKs, reference scripts, installation archives, or laboratory examples. The network layer connects these pieces together through IP addresses, DNS names, hostnames, and routing. If these identities are inconsistent, software that depends on licenses, paths, or remote resources may fail even when the local installation appears correct.

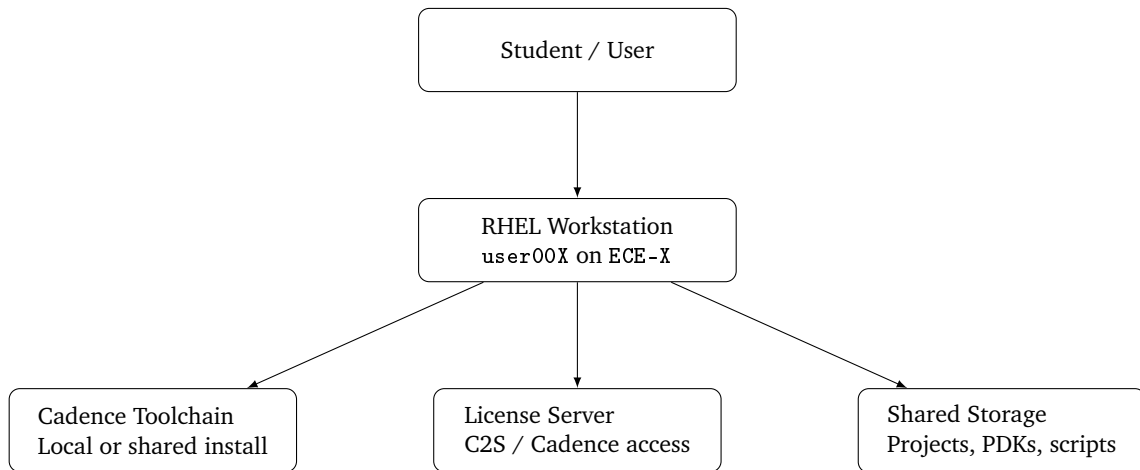


Figure 1.1: High-level view of the laboratory EDA infrastructure

This section is therefore an architectural map, not a detailed installation procedure. It shows what the main pieces are and why they must remain consistent. Later chapters configure these pieces one by one: the operating system is installed, the workstation identity is assigned, the Cadence environment variables are prepared, the license configuration is verified, and the design flow is exercised through RTL design, simulation, synthesis, and physical implementation.

1.6 Workstation Hardware Requirements and Deployment Constraints

Electronic Design Automation tools impose system resource requirements that differ substantially from those of general-purpose desktop software. A typical office productivity environment or software development workstation can operate adequately on modest storage allocations, a few gigabytes of memory, and commodity processor performance. EDA environments do not share these characteristics. The software stacks involved in a complete semiconductor design workflow are architecturally large, computationally intensive, and operationally demanding in ways that must be accounted for at the infrastructure planning stage rather than addressed reactively after deployment.

1.6.1 Storage Requirements

Modern EDA suites consist of numerous interrelated tools, each with its own binaries, libraries, technology files, process design kits, documentation sets, and supporting scripts. A complete installation image for a commercial EDA suite does not occupy gigabytes in the way that a large application suite might. It occupies an order of magnitude more. The Cadence EDA installation deployed in this laboratory environment occupied approximately **480 GB** of storage prior to the creation of any user project data, simulation output, waveform databases, or temporary working files. This figure reflects only the tool installation itself.

In practice, active design work compounds storage consumption considerably. Simulation runs generate large output datasets. Layout databases accumulate across design iterations. Waveform viewers load and cache substantial binary files during interactive sessions. Verification engines write intermediate results to disk throughout their execution. Over the operational lifetime of a workstation, the storage footprint of the design environment grows continuously. Storage planning must therefore account not only for the initial installation image but for a realistic projection of working data volume across the deployment period.

For this reason, workstations intended for sustained EDA use should be provisioned with local storage that provides meaningful headroom beyond the installation baseline. The 480 GB installation figure should be treated as a floor, not a target. Configurations that provision only enough storage to accommodate the initial installation leave no margin for project data, OS overhead, swap allocation, or tool updates, and will require administrative intervention as the environment matures.

1.6.2 Memory Requirements

Semiconductor design workflows routinely involve multiple concurrently active applications. A typical interactive session may involve a layout editor, a schematic capture environment, a circuit simulator, a waveform viewer, and a design rule checking utility operating simultaneously within the same desktop session. Each of these applications maintains its own working set in memory, and several—particularly simulation engines and verification utilities—can expand their memory consumption substantially during active computation.

Insufficient memory allocation produces predictable and disruptive consequences in this context. Operating system memory pressure forces active application data to swap, degrading interactive responsiveness and extending simulation runtimes. In severe cases, the out-of-memory subsystem may terminate running processes, corrupting simulation state and requiring workflow restart. These outcomes are not edge cases in underpowered environments; they are routine operational hazards.

Workstations deployed for full EDA workflows should be provisioned with sufficient physical memory to support concurrent tool operation without inducing swap utilization under normal working conditions. The specific requirement varies by workflow, but **configurations below 16 GB of physical memory are generally unsuitable for sustained multi-tool VLSI design sessions**. Workstations supporting heavier simulation workloads or concurrent multi-user access benefit from higher memory allocations accordingly.

1.6.3 Deployment Methodology: Native Installation versus Virtualization

The manner in which RHEL is deployed on a workstation has direct consequences for the resource availability of the EDA environment. Two deployment models are relevant to this laboratory context: direct installation on dedicated hardware, and virtualized deployment within a hypervisor environment running on an existing host operating system.

In a native installation—whether on a dedicated workstation or in a dual-boot configuration alongside an existing operating system—RHEL has unrestricted access to all physical hardware resources. The full complement of CPU cores, physical memory, storage throughput, and graphics capability is available to the operating system and the EDA tools running within it. This is the preferred deployment model for sustained engineering workflows, as it eliminates the resource-sharing constraints and performance overhead inherent to virtualized environments.

Virtualization, however, represents a practically useful alternative under specific deployment conditions. During the initial workshop phase of this laboratory deployment, Microsoft Hyper-V was used to provision RHEL virtual machines on workstations running existing Windows installations. This approach was selected because many of the available systems contained active Windows environments that could not be immediately repartitioned, and the operational priority was to make the EDA environment accessible within a constrained preparation timeline.

The resource tradeoffs introduced by this approach are significant and should be understood clearly. The reference deployment system carried 32 GB of physical memory. Of this, 16 GB was allocated to the RHEL virtual machine, with the remaining capacity reserved for the Windows host and the memory overhead of the Hyper-V hypervisor layer itself. Under this configuration, the EDA environment operates within a resource envelope that is materially smaller than the physical hardware could otherwise provide. CPU scheduling is mediated by the hypervisor, storage I/O is subject to virtualization overhead, and memory allocation is fixed at provisioning time rather than dynamically available from the full physical pool.

Virtualized deployment is appropriate for introductory workshops, environment familiarization, and short-duration instructional use cases in which the full performance envelope of the EDA tools is not required. It is not the preferred model for production semiconductor design workflows. As workstation preparation schedules permit, systems initially deployed under Hyper-V should be transitioned to native RHEL installations in order to provide the EDA environment with unrestricted access to the underlying hardware resources. All procedures documented in this manual apply to both deployment models unless a specific section explicitly notes otherwise.

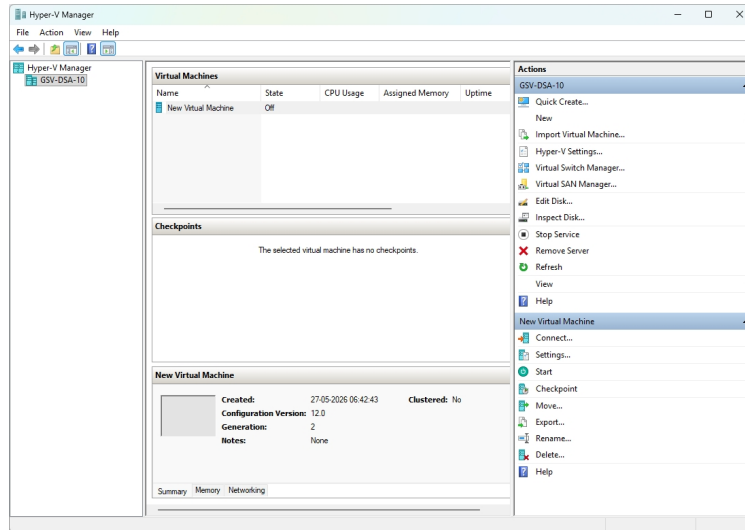


Figure 1.2: Screenshot of Hyper-V

1.7 Summary

This chapter introduced the foundational concepts underlying modern semiconductor EDA infrastructure and established the operational context for the deployment procedures described throughout this manual. The discussion examined the role of Electronic Design Automation within contemporary VLSI workflows, the infrastructure requirements imposed by commercial EDA toolchains, and the reasons enterprise Linux environments have become the standard platform for semiconductor engineering systems.

The chapter further justified the adoption of Red Hat Enterprise Linux as the primary operating system environment for workstation deployment, emphasizing considerations such as long-term stability, vendor certification, maintainability, and compatibility with commercial EDA applications. In addition, the architectural philosophy of the laboratory environment, including standardization, reproducibility, centralized administration, and scalability, was introduced as the guiding framework for all subsequent deployment and maintenance procedures.

Finally, the chapter discussed the hardware and deployment considerations associated with EDA environments, including storage requirements, memory allocation strategies, and the operational tradeoffs between native and virtualized deployment methodologies. These concepts collectively establish the technical foundation required for the operating system installation and workstation provisioning procedures described in the following chapter.

02

RHEL Installation and Setup

Note: This guide assumes you are familiar with Linux environments to a basic degree, are comfortable with the terminal, and have an active internet connection. We will not discuss the entire installation process, only the core infrastructure decisions that need to be made for successful setup of the base system.

That being said however, every decision and command will be discussed in elaborate detail – justifying why it was made, and what it does.

2.1 Installation Planning and Deployment Strategy

Before a workstation is installed, the deployment method must be selected deliberately. The choice is not merely a matter of convenience; it determines how the machine will consume hardware resources, how reliably EDA tools will interact with the operating system, how easily the workstation can be maintained, and how closely the final environment will resemble a production semiconductor design platform. In the context of this manual, RHEL is treated as the operating system baseline for a controlled EDA workstation rather than as a general-purpose desktop installation.

There are two practical ways to deploy RHEL on a laboratory workstation:

1. **Virtualization** – installing RHEL inside a virtual machine hosted by an existing operating system.
2. **Dual booting** – installing RHEL directly on the physical machine alongside an existing operating system by using bootable installation media.

Virtualization is useful for training, experimentation, documentation, and low-risk validation of administrative procedures. It allows the installer to be tested without repartitioning the physical disk and makes it easy to take snapshots before major configuration changes. However, virtual machines introduce an additional abstraction layer between the EDA workload and the hardware. This can limit graphics performance, memory availability, disk throughput, USB access, license-dongle support, and other device-level behavior that may matter during tool deployment or troubleshooting.

Dual booting installs RHEL natively on the workstation hardware. This approach gives the Linux environment direct access to the CPU, memory, graphics adapter, storage devices, and network interfaces, making it the preferred method when the machine is expected to function as a serious EDA workstation. Native installation also more closely matches the assumptions made by commercial EDA vendors, whose installation and support documentation typically targets physical or enterprise-managed Linux systems rather than desktop virtual machines.

The decision made at this stage affects the entire downstream configuration: disk layout, package selection, driver support, user account preparation, license-server connectivity, shared storage mounts, backup strategy, and future system recovery. For that reason, the installation procedure should be selected before any

partitioning or media preparation begins. This chapter documents the installation workflow with attention to university licensing constraints, C2S operational guidelines, and the need for a reproducible workstation baseline.

2.2 Hardware Requirements and Workstation Specifications

A modern semiconductor CAD workstation intended for full-scale Cadence deployment must be provisioned with significantly higher hardware resources than a conventional desktop system. Contemporary EDA workflows involve computationally intensive applications for schematic capture, mixed-signal simulation, physical layout, parasitic extraction, verification, and waveform analysis, many of which operate on large design databases and generate substantial intermediate data during execution. In addition to the operating system itself, a complete Cadence installation together with process design kits (PDKs), technology libraries, simulation environments, temporary build data, and user project directories can consume several hundred gigabytes of storage.

Workstations intended for academic laboratory deployment should therefore prioritize memory capacity, storage performance, multi-core CPU capability, thermal stability, and long-term maintainability over consumer-oriented graphical features. While smaller configurations may support introductory workflows, a poorly provisioned workstation can significantly degrade simulation performance, layout responsiveness, and overall user productivity during large-scale VLSI design tasks.

At the time of writing of this document, the current system available to us had the following specs: *(so you can make a fair estimate of the requirements based on this)*

Component	Detected Specification
Workstation Model	HP Z4 G5 Workstation Desktop PC
System Type	64-bit x86 Workstation Platform
Processor (CPU)	Intel Xeon w3-2423 Processor 6 Physical Cores, 12 Logical Processors Base Frequency: 2.11 GHz
Installed Memory (RAM)	32 GB DDR Memory Installed
Firmware / BIOS Mode	UEFI Boot Mode Enabled
Secure Boot Status	Enabled
Virtualization Support	Hardware Virtualization Enabled Hypervisor and Virtualization-Based Security Active
Motherboard Platform	HP System Board 8962
Operating System	Microsoft Windows 11 Pro for Workstations Build 26200 (64-bit)
Platform Classification	Enterprise Workstation-Class System
Storage Configuration	Not fully listed in exported report Recommended verification of SSD/NVMe capacity before EDA deployment
Graphics Processing	Not listed in current report extract Dedicated GPU verification recommended for OpenGL-based layout workflows

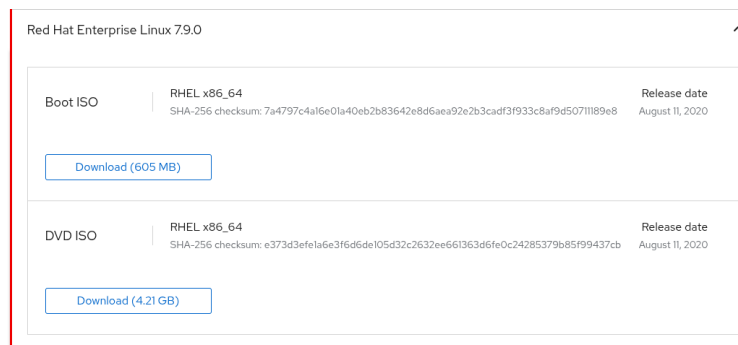
Table 2.1: Host Workstation Specifications Prior to RHEL Deployment

2.3 Obtaining the RHEL Installation Media

The installation workflow begins with obtaining the correct RHEL installation image from Red Hat. A Red Hat Developer account can be used for non-production developer access, subject to Red Hat's current subscription terms and the institution's licensing policy. After signing in, the installer image should be downloaded from the Red Hat Developer portal:

<https://developers.redhat.com/products/rhel/download>

For this workstation deployment, we are using **Red Hat Enterprise Linux 8.7**. This version is selected because many EDA tools and supporting libraries are validated against older enterprise Linux releases, and RHEL 8.7 provides a mature, well-documented platform with broad community and vendor support. In an EDA environment, compatibility and predictable behavior are usually more important than using the newest available operating-system release.



Make sure to download the DVD not the Boot version. The downloaded ISO should match the architecture of the workstation, normally x86_64 for standard laboratory PCs. The version should also match the operating-system baseline approved for the EDA toolchain being deployed. Once the ISO has been downloaded, its integrity should be verified when checksum information is available, and the file should be stored in a documented location so that future workstation rebuilds use the same installation source.

Before writing the ISO to a USB drive or modifying the workstation disk, record the machine's current boot mode, storage configuration, and existing operating-system layout. In particular, confirm whether the system is using UEFI or legacy BIOS boot, whether Secure Boot is enabled, how much unallocated disk space is available, and whether the disk uses GPT or MBR partitioning. These details determine the correct installation path and reduce the risk of inconsistent dual-boot behavior across machines.

2.4 BIOS, UEFI and Secure Boot Configuration

Before installing RHEL, it is important to understand the firmware layer that controls how the machine starts. The firmware initializes hardware, detects bootable devices, and hands control to the operating-system bootloader. Incorrect firmware settings can prevent the installer from booting or can cause the installed operating system to disappear from the boot menu.

BIOS refers to the older firmware interface used by traditional PCs. It performs basic hardware initialization and boots from disks using the legacy boot process, commonly associated with MBR partitioning. Modern workstations generally no longer depend on legacy BIOS, but the term is still commonly used to refer to the system firmware setup screen.

UEFI is the modern replacement for legacy BIOS. It supports GPT partitioning, an EFI System Partition, graphical firmware menus, larger disks, and more flexible boot management. For current workstation deployments, RHEL should normally be installed in UEFI mode so that it matches the boot mode used by Windows and avoids mixed-mode dual-boot problems.

Secure Boot is a UEFI security feature that allows only trusted, signed bootloaders and kernel components to start during the boot process. It helps protect the system from unauthorized boot-time code, but it can also affect Linux installation, third-party drivers, and some virtualization workflows if not configured correctly.

Note: On the HP workstation used in this deployment, the firmware setup screen can be entered by pressing **F2** during startup. On other systems, the key may be different; common alternatives include **F10**, **F12**, **Esc**, or **Delete**, depending on the manufacturer.

Secure Boot will be discussed in more detail in the next section, separately for native dual-boot installation and virtualization-based deployment. The correct setting depends on the installation model, the bootloader path, and whether the deployment requires vendor drivers or kernel-level components that may not be signed for Secure Boot.

2.5 Virtualized RHEL Installation

A virtualized RHEL installation is useful for evaluating the installer, documenting setup procedures, testing package installations, and training users without modifying the physical workstation disk. In this approach, RHEL runs as a guest operating system inside a hypervisor such as VMware Workstation, VirtualBox, or Hyper-V, while Windows remains the host system, making the process low risk because the virtual disk exists only as a file that can be backed up, restored, or removed easily.

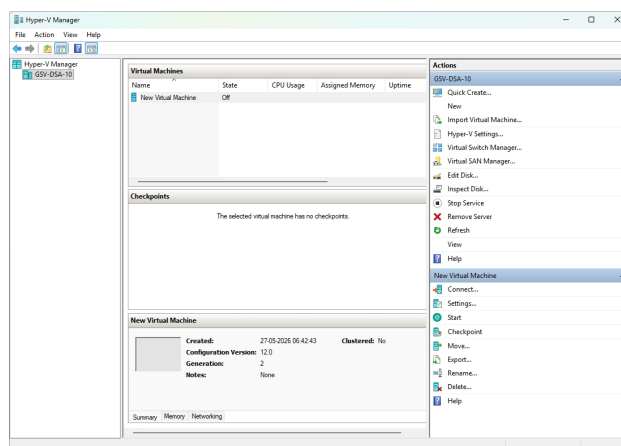
Although virtualization accurately reproduces the RHEL userspace, shell environment, package management, and administrative workflow, it does not fully replicate native hardware behavior, particularly for graphics acceleration, USB passthrough, storage performance, license dongles, and memory-intensive EDA workloads. Therefore, virtualization is best suited for validation, learning, and preliminary testing, while native dual booting remains the preferred choice for production-scale Cadence usage. Before creating the VM, ensure that hardware virtualization support (Intel VT-x or AMD-V/SVM) is enabled in the system firmware.

2.5.1 Preparing the Virtual Machine

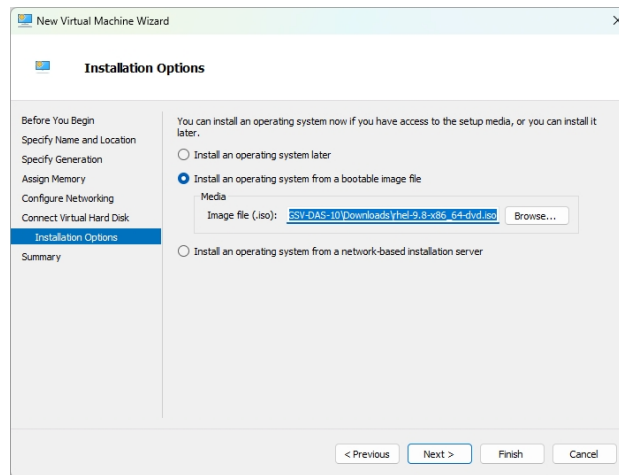
We will need to enable Microsoft Hyper-V for this. Click **Start -> Turn Windows Features On or Off**, scroll till you find the Hyper-V icon and check the box. Restart the system once.

Note: You need Windows Pro or Enterprise to be able to enable Hyper-V. On Windows-Home Hyper-V is not natively supported, but can be installed via a script provided at the end of this chapter.

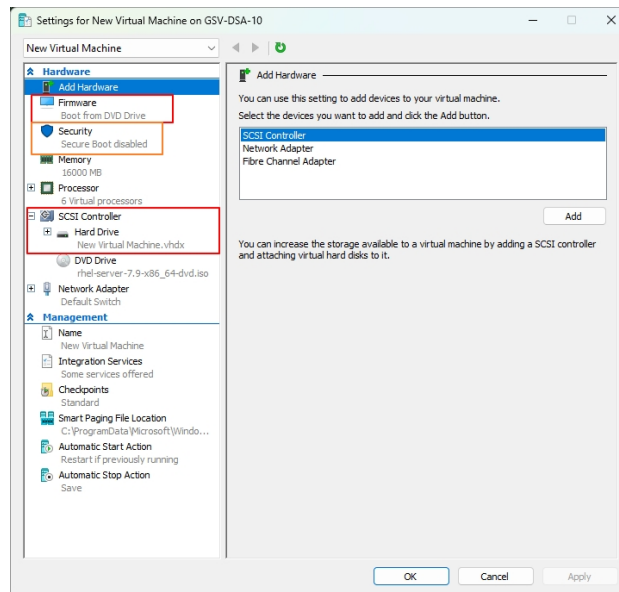
Once you have restarted the system once, open Hyper-V, you will be met with the following screen:



Click **"New" → "Virtual Machine"** and follow the prompts further. Make sure you choose the correct Installation Media – the ISO file we installed earlier.

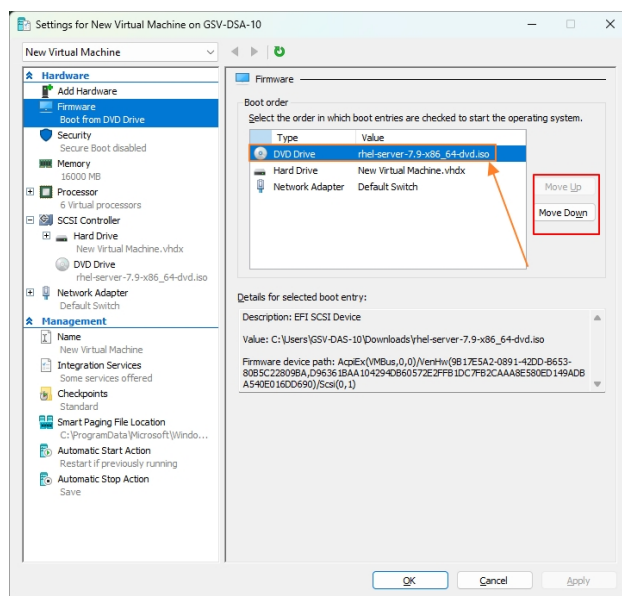


Most of the steps in the prompt are straight-forward. There are a few key configurations you need to keep in mind after setting up the virtual Machine. **Right Click** on the virtual machine, and then click **Settings** to open the following window:

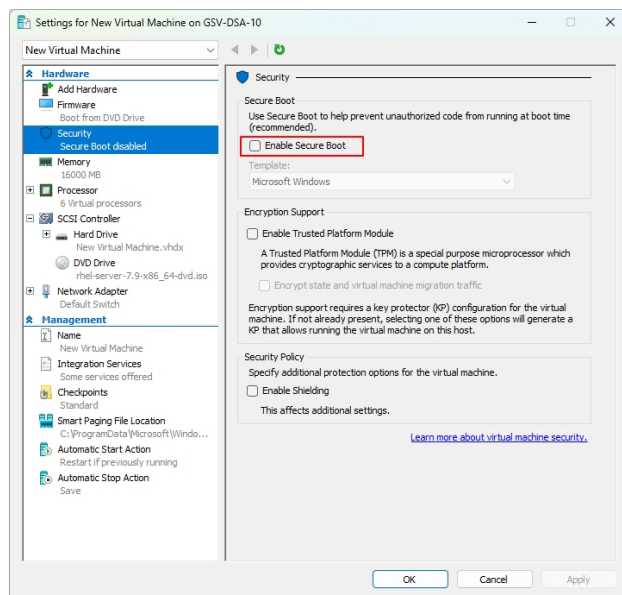


We're interested in only the following three topics, namely:

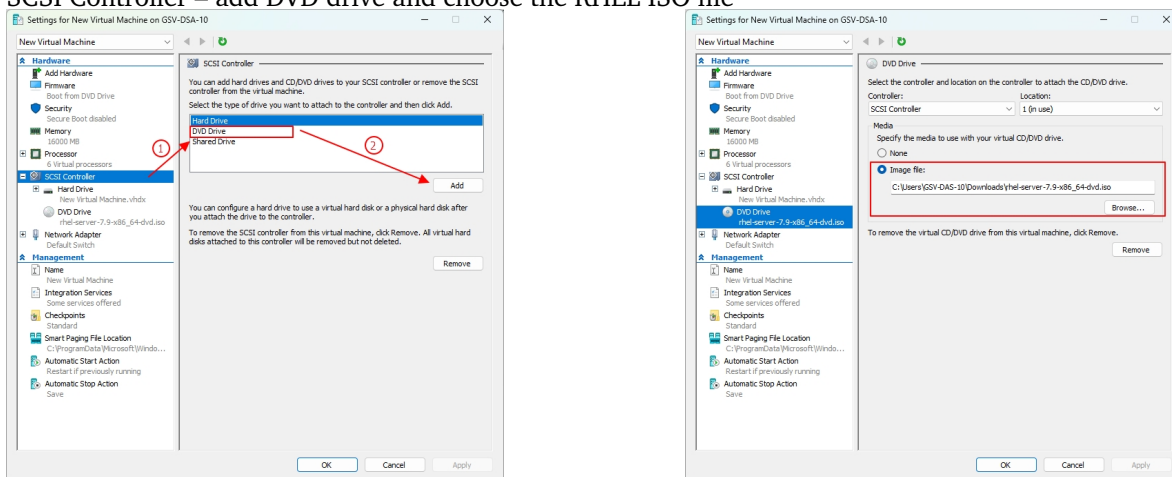
1. Firmware – make sure that DVD drive is at the top of the hierarchy.



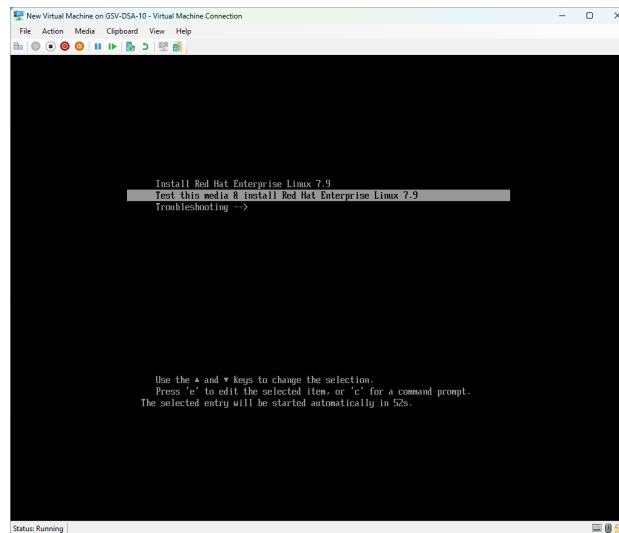
2. Security – disable Secure Boot



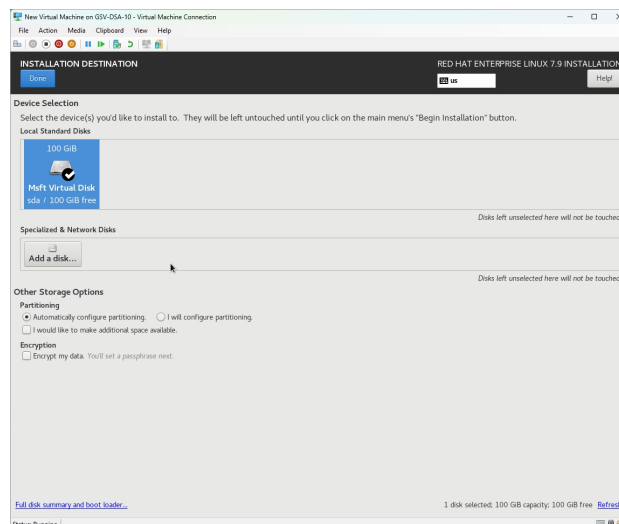
3. SCSI Controller – add DVD drive and choose the RHEL ISO file



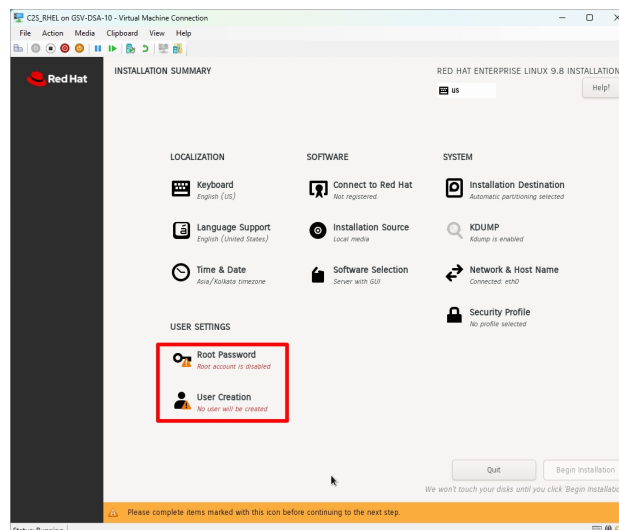
After that close the window and click **Start** and then **Connect**. You should see the following screen:



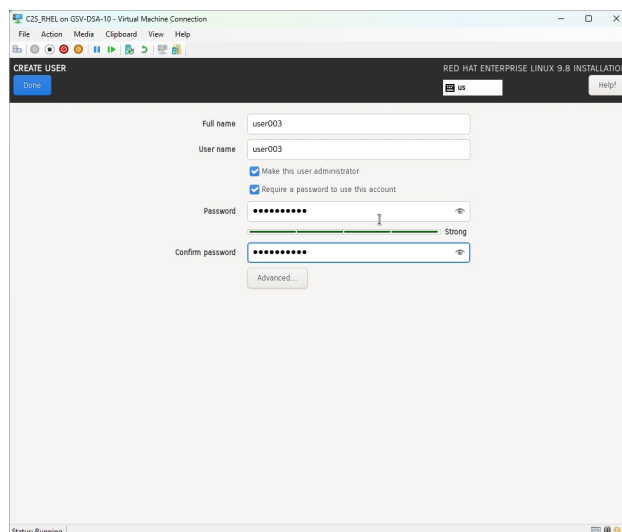
Click "**Install Red Hat Enterprise Linux 8.7**", and follow the setup prompts. For the "Installation Destination" on the Virtual Machine, it should automatically select the Virtual Disk Space (.vhd) you set while setting up the Virtual Machine.



After all this, just one crucial thing is remaining – the username. Click the following cards in the red box, shown on the image on the left below:

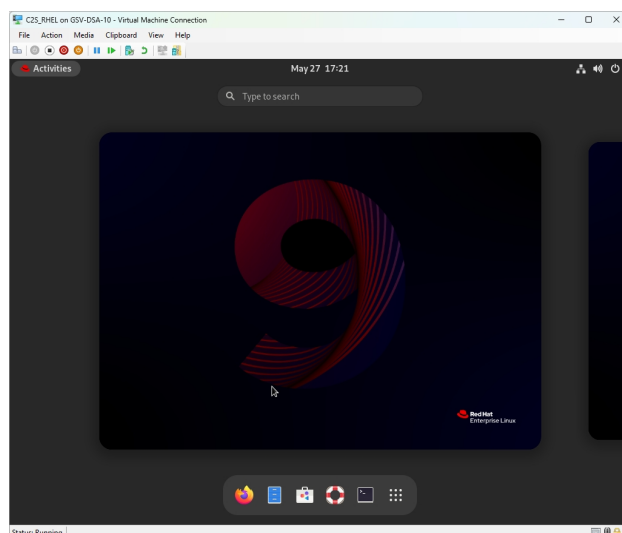


For our use case, we're using a placeholder user named **user003**. This username has been chosen for a specific reason, as will be explained in elaborate detail later in this chapter:



It is generally wise to keep different root and local user passwords, but since we're setting up a workstation for a Laboratory, we keep the passwords the same.

The installation is ready to begin. After the final installation you will be prompted to reboot the virtual Machine. Finally you should see a screen similar to this: (*this is the GNOME desktop environment*)



2.6 Native Dual Booting RHEL 8.7

Dual booting allows RHEL to run directly on the workstation while preserving an existing operating system on a separate partition. This is especially useful in academic laboratories where a machine may need to support both Linux-based EDA workflows and existing instructional or administrative software. The dual-boot model must be planned carefully, because mistakes during partitioning or bootloader configuration can make the existing operating system temporarily unbootable or, in the worst case, destroy data. Important user files should therefore be backed up before installation begins.

2.6.1 Disk Partitioning Strategy

This is a critical stage in the overall system deployment pipeline and must be approached with precision and caution. Any misconfiguration at this level can result in irreversible data loss, including corruption or

removal of existing operating systems such as Windows.

Disk partitioning is the process of logically dividing a physical storage device (SSD or HDD) into multiple independent regions, known as partitions. Each partition behaves as a separate storage volume and is assigned a specific role within the operating system architecture.

In Linux-based system design, partitioning is not arbitrary. It is a structured layout intended to separate system-critical components, improve stability, simplify recovery, and support long-term maintainability of the environment.

A standard Linux workstation deployment typically includes the following core partitions:

1. EFI Partition – *mounted at /boot/efi/*

This is a mandatory boot partition on UEFI-based systems. It stores bootloaders, firmware interfaces, and essential startup binaries required for system initialization. Without this partition, the system cannot boot in a modern UEFI environment.

2. Root Partition – *mounted at /*

This is the primary filesystem hierarchy. It contains the operating system, installed packages, system binaries, configuration files, and most runtime environments. It effectively forms the core of the Linux installation.

3. Swap Partition

This is a dedicated memory extension area used when physical RAM is fully utilized. It supports memory paging, system hibernation (if enabled), and prevents abrupt failures under memory pressure. While modern systems often use swap files, a dedicated swap partition is preferred in controlled workstation environments for predictability and performance consistency.

You may need to manually create a partition based on your system configuration. Press **Windows Key** and then type **Disk Management**. You should see the following:

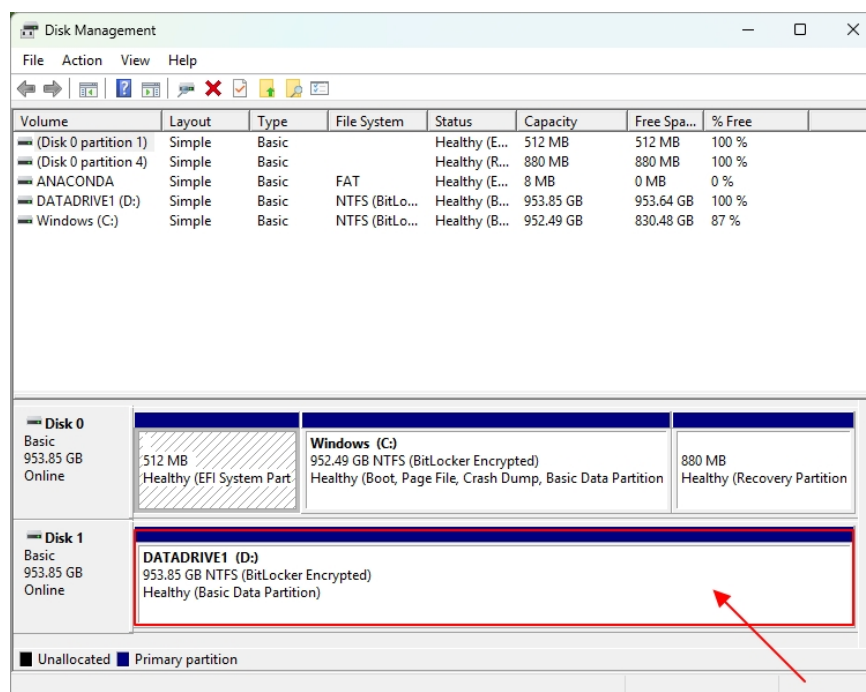


Figure 2.1: Disk Management

From the screenshot, we already have a separate disk — **Disk 1**, with a capacity of **1 TB**. In this case, we can directly use the entire disk without creating additional partitions. However, in the case of a single shared disk, it may be necessary to shrink existing unallocated space in order to create a new partition for installation or data separation.

A detailed walkthrough of this process can be found here: <https://www.youtube.com/watch?v=Wv0Id5CRmw8>.

For this demonstration, we will skip the partitioning step. The key point of interest here is **disk nomenclature in Linux**. Unlike Windows, which uses drive letters such as C: or D:, Linux represents storage devices using a structured naming convention. For NVMe SSDs, the naming format is:

$$\text{nvmeXnYpZ}$$

where:

- X denotes the NVMe controller number (i.e., the physical interface index),
- Y denotes the namespace number (a firmware-defined logical storage unit),
- Z denotes the partition number (a software-defined subdivision of the namespace).

Thus, from the observed system output, we can infer the following mappings:

Disk 0	→	nvme0n1
Disk 1	→	nvme1n1

Furthermore, partitions within these disks are represented as extensions of the namespace identifier. For example:

Disk 0, Partition 1	→	nvme0n1p1
Disk 0, Partition 2	→	nvme0n1p2
Disk 1, Partition 1	→	nvme1n1p1

Here, the namespace (nY) represents a firmware-defined logical block device exposed by the NVMe controller, while partitions (pZ) are operating system-defined subdivisions within that namespace used for filesystem organization and data separation.

Note: The important thing here to remember the name of the disk we want to install RHEL on. In our case, its **nvme1n1**. The mount-point will be at **/dev/nvme1n1**

2.6.2 Creating the Installation Media

For dual-booting, we need to create something known as a "**bootable-drive**". The process is straight-forward. All you need are a USB with atleast 8GB storage capacity, and the ISO we downloaded earlier. To create a bootable-drive, you need a flashing tool.

For Windows, the most popular choice is Rufus; can be installed here: <https://rufus.ie/en/>. If you're on Linux, the most popular choice is Balena Etcher; can be installed here: <https://etcher.balena.io/>.

Plug in your USB and make sure it's mounted. Rufus automatically detects it.

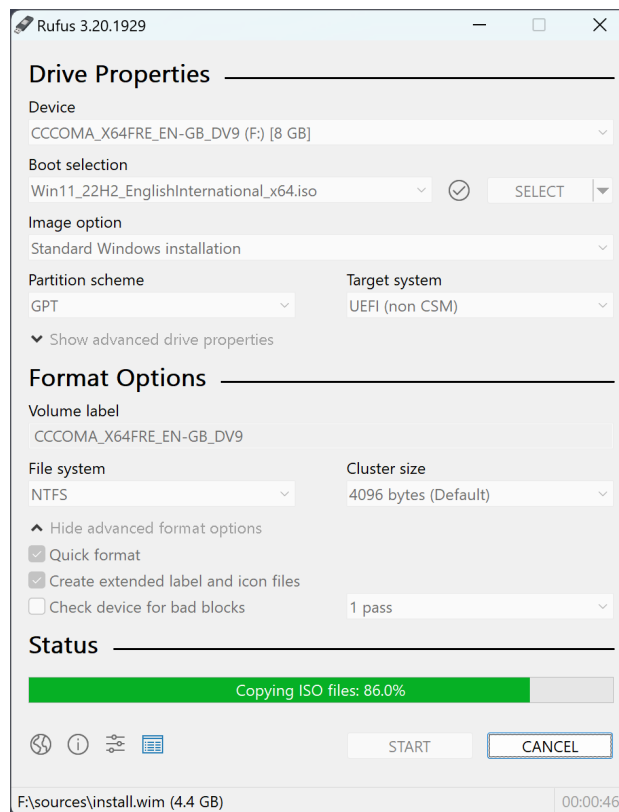


Figure 2.2: Example Screenshot of Rufus

This is a screenshot of the Rufus Tool. The *Device* is usually automatically selected as your USB drive; if not, you can manually select it. Set the *Boot selection* to the RHEL 8.7 ISO we just downloaded. For a modern workstation, use **GPT** as the partition scheme and **UEFI** as the target system. GPT is preferred because current Windows installations and enterprise workstations normally boot in UEFI mode, and using the same boot mode for both Windows and RHEL avoids mixed-mode bootloader issues. Use **MBR** only if the machine is confirmed to be using legacy BIOS boot. On Windows, this can be checked from **System Information** by looking at **BIOS Mode**: choose UEFI if it says *UEFI*, and choose legacy/MBR only if it says *Legacy*.

For the file system, keep the Rufus default unless there is a specific reason to change it. **FAT32** is the safest choice for UEFI boot media because it is directly supported by UEFI firmware; **NTFS** may be used if Rufus requires it because of ISO file-size constraints. Do not choose **EXT4** for the installation USB, since it is not reliably readable by PC firmware during boot and is meant for Linux partitions after installation. Leave the cluster size at **Default**, typically **4096 bytes**, since changing it provides no practical benefit for this installer.

Once this completes, the USB is now ready for installation. Reboot your system, and *enter the Boot Menu* by pressing the relevant key. In our case, we're working on an HP workstation, so we're pressing **F9**.

2.6.3 Installation Destination Stage

This is the main step for dual booting, and usually the place where most novices mess up. Most of the steps are the same as the virtualization setup. The only thing we're going to have to do differently here is selecting the disk where the base system will be installed. Click on the "**Installation Destination**" option from the image below, and select the disk we chose in Subsection 2.6.1 – *nvme1n1*

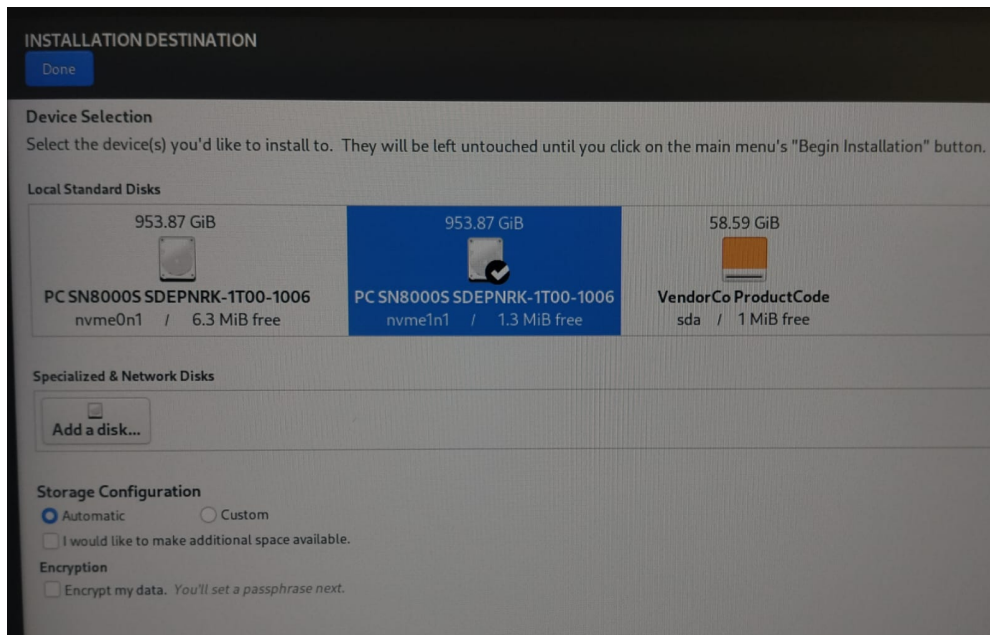


Figure 2.3: Installation Media Selection

Once the base system installs, you should remove the installation media and wait for the system to reboot.

2.7 Chapter Summary

This chapter established the preparation steps required to deploy RHEL 8.7 for Cadence and other EDA workflows. It compared virtualization and native dual booting, noting that virtualization is useful for testing and training, while native installation provides direct access to hardware resources and is therefore preferred for production EDA environments. The chapter also outlined the workstation hardware requirements, documented the target HP workstation, and explained the selection of the RHEL 8.7 DVD installation image for compatibility with enterprise toolchains.

The chapter then reviewed the firmware and boot environment, including the differences between BIOS, UEFI, and Secure Boot. Emphasis was placed on maintaining a consistent boot mode between Windows and RHEL to avoid bootloader and recovery issues. For virtualized deployments, a Hyper-V-based installation procedure was presented, including virtual machine creation, ISO attachment, firmware configuration, Secure Boot settings, and completion of the base operating system installation.

For native dual-boot deployment, the chapter introduced the required Linux partition layout and explained the mapping between Windows disk numbering and Linux NVMe device names to prevent accidental data loss. Finally, a bootable RHEL installation USB was created using Rufus with GPT and UEFI settings appropriate for the target workstation. At this stage, the system is prepared to boot the installer and proceed to disk selection and operating system installation.

03

Post Install Configurations

System identity, licensing, and workstation preparation

In this chapter we will discuss the post-installation configurations required for the deployed workstations. This includes setting up the appropriate hostname and username nomenclature aligned with the license provided by C2S, preparing the system identity used by the Cadence environment, and installing the relevant software packages required for the Cadence Suite.

3.1 Username/Hostname Configuration

After the base RHEL installation is complete, the first administrative task is to standardize the identity of each workstation. In an EDA laboratory environment, usernames and hostnames should not be chosen casually. They are used by the operating system, network services, license checks, remote administration scripts, shared storage mounts, and Cadence startup environments. If different machines follow different naming patterns, the environment becomes difficult to audit and errors in license or path configuration become much harder to diagnose.

According to the license our university has received, the workstation identity should follow the following naming convention:

`user0_X @ ECE-X.gsv.ac.in`

Here, `user0X` represents the local Linux username assigned to the workstation user, while `ECE-X.gsv.ac.in` represents the fully qualified hostname assigned to that workstation on the department network. The value of `X` should be replaced by the workstation number or seat number allocated by the laboratory administrator. For example, workstation 3 may use the username `user003` and the hostname `ECE-3.gsv.ac.in`. The important point is that the same numbering scheme must be used consistently across the user account, hostname records, license configuration, and any administrative documentation.

This convention gives every machine a predictable identity. A system administrator can immediately identify which physical workstation corresponds to which Linux account and hostname, and the Cadence license environment can be configured against a stable naming scheme. Before installing or launching Cadence tools, confirm that the selected username and hostname match the C2S license documentation and the laboratory workstation inventory. If a machine is renamed later, related configuration files, license references, and shell startup scripts may also need to be updated.

At the time of writing this document, there are officially upto 100 registered workstations for GSV students under the C2S License. The first 70 systems are reserved for the ECE department, while the remaining 30 are reserved for EE department.

This very first priority is ensuring the groupname and username/hostnames are as per the respective workstations. File ownership depends heavily on the groupname and GID (group ID). In Linux, the default file permissions and ownership records are verified using GID and UID (user ID), hence ensuring they agree,

resolves problems down the line.

On a fresh install, usually the default group ID is set to **1000** for the default user. In our case it is **user003**. You can verify this with the following commands:

```
id user003 && whoami && hostname
```

The output should display the expected username, User ID (UID), group name, Group ID (GID), and hostname associated with the workstation. These values should be verified before proceeding with the Cadence installation, as several configuration files and environment settings may reference the user account, host identity, or group ownership. If any of these values are incorrect, they should be corrected prior to continuing with the software installation process.

Once the workstation identity has been verified, the next step is to refresh the package metadata and update the operating system packages. On RHEL 9.8, this is typically performed using the `dnf` package manager. The following command updates all installed packages to their latest available versions:

```
sudo dnf update -y
```

After the update process is complete, the workstation should be rebooted if any kernel, driver, or system library updates have been installed. This ensures that the running operating system matches the updated package state before any Cadence-specific dependencies or environment configurations are applied.

The primary configuration tasks at this stage involve setting up the required environment variables, configuring executable search paths, and ensuring that the appropriate file permissions are assigned to the intended user account. Since multiple workstations must be configured with identical software environments, a system imaging and cloning approach is employed using Clonezilla.

After a workstation image has been cloned using Clonezilla, the target machine inherits several machine-specific attributes from the source workstation, including the username, hostname, home directory structure, and file ownership information. These parameters must be customized before the workstation can be deployed for regular use.

Traditionally, this process would require the administrator to manually perform the following tasks:

- Renaming the user account
- Renaming the primary group
- Updating the user's display name
- Updating the hostname
- Updating `/etc/hostname`
- Updating `/etc/hosts`
- Correcting ownership of user directories

Performing these operations manually on every cloned workstation is repetitive and increases the likelihood of configuration errors. To streamline the deployment process and ensure consistency across all laboratory systems, these tasks have been automated through a dedicated C-shell (`csh`) configuration script. The script prompts the administrator for the required workstation-specific parameters and automatically applies the necessary user, group, hostname, and ownership modifications, thereby reducing deployment time and minimizing the risk of human error.

The full script is present in Appendix A

3.2 Installing and Configuring Cadence

3.2.1 Cadence Directory Structure

Cadence expects everything to be present in the `/home/installs` directory, so we need to make that:

```
sudo mkdir -p /home/installs
```

This is the place where we install all the tools in the Cadence Suite and unzip them.

For a laboratory deployment, the `/home/installs` directory should be treated as a centralized local software repository rather than as an arbitrary extraction location. All Cadence releases, auxiliary setup files, and vendor-provided directory trees should be placed under this common root so that every workstation follows the same filesystem layout. This standardization is important for system imaging, recovery, and later maintenance. If one workstation stores the Cadence tree in a different location, the shell startup file, environment variables, license diagnostics, and teaching scripts may need to be modified only for that machine, defeating the purpose of a common laboratory image.

The Cadence installation archives used for the C2S laboratory setup are obtained from the repository provided for the deployment. Administrators should download the packages from the following source and preserve the original archive structure supplied there:

```
https://entupletech-my.sharepoint.com/personal/cadence_support_entuple_com/_layouts/15/onedrive.aspx?id=%2Fpersonal%2Fcadence%5Fsupport%5Fentuple%5Fcom%2FDocuments%2FDATA%2FCadence%5FRHEL%5F8%5Fversion%2FIndividual%5Flinks&ga=1
```

In normal workstation preparation, the archive is first downloaded into the user's `~/Downloads` directory. This keeps the downloaded source material separate from the final installation location and allows the administrator to inspect the archive before extraction. After verification, the archive must be extracted directly into `/home/installs/`. No additional wrapper directory should be created during extraction. The contents of the archive should land directly inside `/home/installs/` so that the Cadence directory hierarchy expected by the later environment-variable, C-shell initialization, and license setup steps is preserved.

The exact archive name may differ depending on the release downloaded from the repository. The following example therefore uses a clearly marked placeholder for the archive filename:

```
cd ~/Downloads
ls -lh

# Replace this with the actual archive downloaded from the repository.
ls -lh <CADENCE_ARCHIVE_NAME>

sudo tar -xf <CADENCE_ARCHIVE_NAME> -C /home/installs/

ls -lh /home/installs
find /home/installs -maxdepth 2 -type d | sort
```

If the downloaded file is provided in a different compressed format, the extraction command should be adjusted accordingly while keeping the destination unchanged. For example, a `.zip` archive should still be extracted with `/home/installs/` as the target directory. The administrative rule remains the same: do not create an extra directory layer between `/home/installs/` and the Cadence tree supplied by the archive.

Preserving the original hierarchy is critical because subsequent configuration steps refer to predictable paths below `/home/installs`. The `.cshrc` file, Cadence environment variables, executable search path, runtime library path, and license diagnostic commands all assume that the installation tree has not been rearranged

manually. A small change made during extraction can later appear as a shell configuration failure or a license startup problem, even though the root cause is simply that the expected directory structure was not preserved. For this reason, the extraction layout should be verified immediately after unpacking the archive and before any environment variables are written.

3.2.2 Environment Variables

Before launching any Cadence tool, the user's shell environment must point to the correct installation directories and license server. If a working `.cshrc` file is already present and contains the required paths, it does not need to be recreated. The important requirement is that the active shell can locate the Cadence executables, libraries, setup files, and license configuration.

The following values should be checked for every workstation:

- The Cadence installation root under `/home/installs`
- The executable paths for Cadence tools, usually the `bin` and `tools/bin` directories inside the Cadence installation
- The runtime library path required by the installed Cadence release
- The license variables, such as `CDS_LIC_FILE` and `LM_LICENSE_FILE`, pointing to the correct license server and port
- Any release-specific setup paths required by the installed Cadence package

The complete C-shell startup configuration is provided in the appendix: B. After confirming or updating the startup file, log out and log back in, or source the file manually, and verify the active environment using:

```
terminal
echo $CDS_INST_DIR
echo $CDS_LIC_FILE
which genus
```

If `which genus` does not return a path under the Cadence installation directory, the executable path has not been configured correctly. The environment variables should be corrected before continuing with license checks or tool validation.

3.2.3 C-Shell Configuration

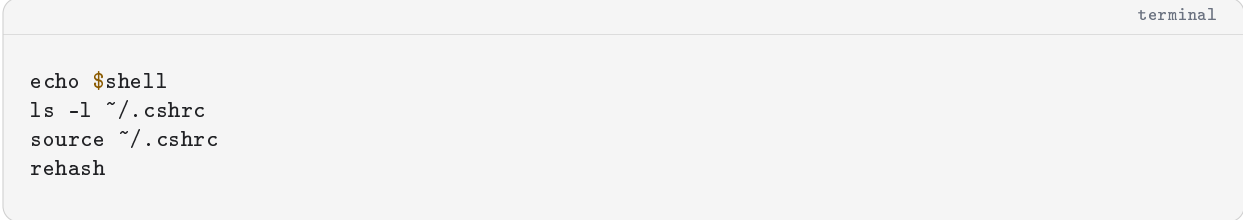
The C-shell startup file, usually stored as `~/.cshrc`, is one of the most important configuration files in a Cadence workstation deployment. It is read whenever an interactive C-shell session is started, and it defines the environment that the user receives before any EDA tool is launched. In a single-user Linux installation this file may only contain a few aliases or display preferences, but in an EDA laboratory it becomes part of the workstation infrastructure. It controls which tool release is visible, which license server is contacted, which library paths are searched, and whether the shell behaves consistently across cloned systems.

Cadence installations have traditionally relied on C-shell based initialization because many vendor-provided setup scripts, examples, and legacy flows assume `csh` or `tcsh` syntax. Even when the operating system default shell is Bash, laboratory users may still be instructed to start Cadence tools from a C-shell session so that the same environment logic is applied on every workstation. This reduces variation between machines and prevents accidental differences between users who may otherwise maintain separate Bash, Zsh, or graphical terminal profiles. For a C2S laboratory, where the same toolchain must be reproduced across many student systems, this consistency is more important than personal shell preference.

The `.cshrc` file should therefore be treated as a controlled configuration file. Administrators should avoid scattering Cadence variables across multiple unrelated startup files, because such a layout makes troubleshooting difficult after cloning or system repair. A maintainable configuration should clearly define the

installation root, append the required executable directories to the shell path, define any required runtime library locations, and set the licensing variables used by Cadence and FlexLM. If several Cadence releases are installed under `/home/installs`, the selected release should be documented in the file so that future administrators can identify which version is active without inspecting every directory manually.

A minimal administrative check is to confirm that the user account is actually reading the expected C-shell startup file. This should be tested from the same account that will run the tools, not only from the root account, because file ownership, home directory changes, and cloned user profiles can easily produce different results for different users. The following commands may be used after logging in as the workstation user:

A terminal window with a light gray background and a title bar that says "terminal". It contains four lines of text:

```
echo $shell  
ls -l ~/.cshrc  
source ~/.cshrc  
rehash
```

```
echo $shell  
ls -l ~/.cshrc  
source ~/.cshrc  
rehash
```

The `source` command reloads the startup file in the current shell, while `rehash` refreshes the C-shell command lookup table after new executable directories have been added to the path. This is useful during installation because the administrator can edit the startup file and test it immediately without logging out after every change. However, final validation should still be performed in a new login session, because that confirms that the settings are loaded automatically and do not depend on manual commands typed during installation.

The C-shell configuration should not contain workstation-specific values unless those values are intentionally different for each machine. The tool installation path and license configuration should normally be identical across all cloned workstations, while the username and hostname are handled separately during the system identity configuration described earlier in this chapter. This separation improves maintainability: a cloned machine can receive its own hostname and user identity without requiring a different Cadence startup file for every seat in the laboratory.

3.2.4 License Configuration

Cadence tools are protected by network licensing, commonly implemented through the FlexLM licensing system. In this model, the workstation does not independently decide whether a tool may run. Instead, when a Cadence executable starts, it contacts the configured license server and requests the required license feature. The server checks whether the feature exists, whether the request is allowed, and whether enough license tokens are available. Only after this exchange succeeds does the tool proceed to the normal graphical or command-line startup sequence.

License configuration is therefore a core infrastructure dependency, not a cosmetic environment setting. A workstation may have a complete Cadence installation and a correct executable path, but the tools will still fail to launch if the license variable is missing, the server hostname cannot be resolved, the port is incorrect, or the network path to the server is unavailable. In a teaching laboratory this can affect many users at the same time, so license settings should be kept centralized, documented, and reproducible. The same license configuration should be applied to every workstation unless the C2S license documentation explicitly requires separate values for separate systems.

The two variables most commonly involved are `CDS_LIC_FILE` and `LM_LICENSE_FILE`. The Cadence-specific variable points Cadence tools to the license source, while `LM_LICENSE_FILE` is understood by FlexLM-based utilities and may also be used by other licensed software. The value normally follows the form `PORT@LICENSE_SERVER`, where `PORT` is the FlexLM server port and `LICENSE_SERVER` is the hostname or IP address of the license server. The actual value must be taken from the C2S license documentation or from the laboratory license administrator. If the value is not yet known during documentation preparation, it should be left as a clearly marked placeholder rather than replaced by an assumed address:

```
csh

setenv CDS_LIC_FILE <PORT>@<LICENSE_SERVER_HOSTNAME>
setenv LM_LICENSE_FILE $CDS_LIC_FILE
```

Hostname resolution should be checked before blaming the Cadence installation itself. If the license variable uses a hostname, that hostname must resolve through DNS or through a controlled entry in `/etc/hosts`. If the laboratory network is segmented, administrators should also confirm that the workstation can reach the license server from the student VLAN or laboratory subnet. Using an IP address can avoid name resolution problems, but it also makes future server migration less transparent. For maintainability, a stable hostname is usually preferable when the network infrastructure supports it correctly.

Licensing failures commonly occur because the startup file was not sourced, the wrong shell was used, the license server name was mistyped, the FlexLM port was changed, or a firewall blocked the connection. Another frequent cause is a mismatch between the installed Cadence release and the features available in the license file. For this reason, license validation should be performed after every installation, after every workstation clone, and after any license server maintenance window. The following checks are useful when the FlexLM utility is available in the active environment:

```
terminal

echo $CDS_LIC_FILE
echo $LM_LICENSE_FILE
lmutil lmstat -a -c $CDS_LIC_FILE
```

If `lmutil` is not available directly in the shell path, the administrator should use the copy provided by the installed Cadence release, or replace the command with the site-approved license diagnostic utility. The important point is not the exact utility name, but the validation principle: the workstation must be able to contact the same license source that Cadence will use at runtime.

3.2.5 Verifying Installation

Verification should be performed systematically before a workstation is released for student use. The goal is not merely to prove that one command opens once, but to confirm that the deployed machine matches the intended laboratory baseline. This includes the operating system identity, the Cadence installation directory, the C-shell startup configuration, the executable search path, the runtime library environment, and access to the license server. A structured validation step is especially important after Clonezilla deployment, because cloned machines can inherit paths and user configuration from the source image while still requiring workstation-specific identity changes.

The first validation step is to check that the environment visible to the user is the same environment described in the configuration files. This must be done from a fresh terminal session as the intended user account. Administrators should avoid relying only on commands run immediately after manual sourcing, because that can hide startup errors that will reappear when students log in normally. The following commands provide a basic view of the active shell and Cadence environment:

```
terminal

whoami
hostname
echo $shell
echo $CDS_INST_DIR
echo $CDS_LIC_FILE
echo $LM_LICENSE_FILE
```

The values should match the workstation identity convention and the Cadence configuration approved for the laboratory. Empty output for a required variable usually indicates that the startup file was not read, the user is not running the expected shell, or the variable was defined in a file that is not part of the current login

sequence. Incorrect output is more serious than empty output, because it may point the user to an outdated release or to an obsolete license server without making the failure obvious at first.

The next step is to confirm executable visibility. Cadence tools are large application suites, and their front-end commands must be reachable through the C-shell path. If the shell cannot find the executable, the installation may still exist on disk, but the user environment is not ready. The administrator should check both command discovery and version reporting where the tool supports it:

```
terminal

which genus
which xrun
which innovus

genus -version
xrun -version
innovus -version
```

Not every laboratory installation will include every Cadence product. A missing command is therefore not automatically an error if that product was not installed or was not included in the C2S entitlement. However, every tool expected for the course flow should resolve to a path under the approved Cadence installation directory, and its version should correspond to the documented release. If different machines report different versions, the clone image or startup file should be reviewed before the laboratory is used for instruction.

License access should be validated after executable visibility, because a tool may be installed correctly but still fail during feature checkout. If FlexLM diagnostics are available, the license server can be queried before opening a full graphical application:

```
terminal

lmutil lmstat -a -c $CDS_LIC_FILE
```

Finally, the administrator should perform a practical launch test using at least one graphical tool and one command-line or batch-capable tool from the expected design flow. This catches problems that simple path checks may miss, such as missing shared libraries, display forwarding issues, incompatible system packages, or permission problems in the user's working directory. The workstation should only be considered ready when these checks succeed from the normal student account without requiring root privileges.

3.2.6 Launching Cadence Tools

Cadence tools should be launched from a prepared user environment, not from an arbitrary terminal state. In normal laboratory use this means that the user logs in to the workstation, opens a C-shell-compatible terminal session, and starts the required tool from a project working directory. The working directory matters because many Cadence applications create local run files, logs, lock files, simulation outputs, or design database directories during startup. Launching tools from a temporary or root-owned directory can create permission problems that are difficult for students to diagnose later.

During the first execution, the tool performs several checks before the user reaches the main interface. It reads the shell environment, locates the installation tree, loads shared libraries, checks release-specific setup files, contacts the license server, and initializes user configuration data. A delay during the first launch is therefore not unusual, especially on a newly cloned workstation or after a user profile has been created. However, repeated failures at this stage usually indicate an infrastructure problem: an incorrect path, a missing library, an unavailable license feature, or an environment mismatch between the selected release and the active shell.

A typical launch test for the synthesis environment is performed with `genus`. The administrator should run it from a writable directory owned by the user, observe whether the application starts without license errors, and then close it normally after the startup sequence is complete:

terminal

```
mkdir -p ~/cadence_test  
cd ~/cadence_test  
genus &
```

Successful startup confirms more than the presence of the `genus` executable. It confirms that the shell path is active, the required runtime libraries can be loaded, the license server is reachable, and the release selected in the shell matches the installation tree on disk. If the tool reports license warnings or missing setup files, the workstation should not be handed over as ready until those messages are understood. Warning messages that appear harmless during installation often become recurring support issues once a full class begins using the machines.

Digital design and implementation tools may also be checked from the command line. Version queries are useful for confirming the active release without starting a full project, while a later course-specific flow can be used for deeper validation. The following commands provide a lightweight launch-level check for commonly used Cadence tools, subject to the products actually installed under the C2S deployment:

terminal

```
xrun -version  
genus -version  
innovus -version
```

A successful launch or version response should be recorded as part of the workstation handover checklist. In a laboratory setting, this record helps distinguish between installation problems and later user-level project errors. Once the major tools start from the normal user account, the expected license source is reachable, and the reported versions match the documented release, the Cadence environment can be considered ready for instructional use.

Part II

RTL DESIGN FUNDAMENTALS

Introduction to Verilog and basic RTL design

04

Introduction to Verilog

System identity, licensing, and workstation preparation

4.1 Why Hardware Description Languages Exist

As digital systems grew beyond small collections of gates, schematic-only design became unsuitable for serious integrated-circuit development. A modern chip must be described, simulated, revised, and verified long before it reaches physical implementation, and this requires a representation that EDA tools can process consistently. **Hardware Description Languages (HDLs)** provide this representation. They allow engineers to describe logic, registers, interconnections, timing intent, and hierarchy in text, while still representing hardware rather than software instructions executed by a processor.

Verilog is one of the primary HDLs used for digital design and verification. It can describe simple combinational circuits, sequential elements, counters, multiplexers, and complete RTL blocks that later pass through simulation, synthesis, and implementation. Its main value in this flow is that the same design can be organized hierarchically, tested before fabrication, and reused across larger systems. This chapter introduces the basic Verilog program structure, modules, ports, data types, constants, and operators needed for the RTL design and verification work used in later chapters.

4.2 Verilog Program Structure

A Verilog source file is organized around one or more modules, and each module acts as a self-contained hardware block with its own ports, declarations, and internal logic. This structure is not a stylistic convention; it is what allows EDA tools to parse the design, elaborate the hierarchy, and determine which parts of the description are meant to behave as combinational logic, sequential logic, or hierarchical instances of other blocks. A complete Verilog design therefore begins with the module header, continues with the declaration of ports and internal objects, and ends with one or more behavioral or structural descriptions that define how the block operates.

The general pattern is consistent across small laboratory examples and larger RTL blocks. The module header names the design unit and lists its external interface. The body then defines whether each port is an input, output, or inout, assigns widths where required, declares nets and variables, and introduces continuous assignments or procedural blocks. A module may also contain parameters, submodule instances, tasks, functions, and generate constructs when the design grows in complexity. The following skeleton shows the layout used throughout this manual:

```
module module_name (port_list);  
    // port directions and widths  
    // internal declarations
```

verilog

```
// continuous assignments
// procedural blocks
endmodule
```

In practice, this structure allows the same description to be read by both simulator and synthesis tools. Simulation uses the procedural and concurrent statements to model behavior over time, while synthesis extracts the logic implied by the supported constructs. For that reason, the organization of the code matters as much as the syntax itself. A clean module layout makes a design easier to review, easier to debug, and easier to hand over to another engineer or laboratory administrator later in the flow.

4.3 Modules

The module is the basic unit of composition in Verilog. Every hardware description is built from one or more modules, whether the design represents a simple gate-level block or a complete subsystem with multiple submodules. A module may be reused many times within a larger design, which is why it is useful to think of it as a hardware component rather than as a software procedure. When one module instantiates another, the relationship is fixed at elaboration time; the resulting hierarchy mirrors the structure of the hardware that will be simulated or synthesized.

The module body can include instances of other modules, primitive gates, continuous assignments, tasks, functions, and procedural blocks. A small module may contain only a few lines of declaration and a single assignment, while a larger module may include several nested blocks and many internal signals. The important point is that the interface remains explicit. For example, a full adder can be described with only a few statements:

```
verilog

module full_adder(A, B, Cin, S, Cout);
    input A, B, Cin;
    output S, Cout;

    assign {Cout, S} = A + B + Cin;
endmodule
```

The same block may later be reused as part of a larger arithmetic unit without changing its interface. This reuse is one of the practical reasons Verilog became central to RTL design. It allows an engineer to define a small verified building block once, then connect that block into a wider system by instantiating it repeatedly. The module boundary therefore serves both as a design boundary and as a documentation boundary.

4.4 Ports

Ports define the external interface of a module. They are the connection points through which the module exchanges signals with the rest of the design, and they must be declared explicitly as input, output, or inout. A port list alone is not enough to define the signal type; the direction and, when necessary, the data type and bit range must also be stated inside the module body. In many cases a port is treated as a wire by default, but wider buses and registered outputs require additional declaration.

The port list becomes especially important when a module is parameterized. Widths can be described using parameters so that the same source file may be reused for different data widths without rewriting the logic. This is common in laboratory designs, where a small example may later be scaled to a wider bus for experimentation. The syntax below shows the basic pattern:

```
verilog

module module_name
    #(parameter WORD = 32)
```

```

(addr, sum, a, b, sel, data_bus);

input  [WORD-1:0] addr;
output reg signed [32:1] sum;
input  a, b, sel;
input  [0:15] data_bus;
endmodule

```

The declaration order matters when the design is read by tools and by human reviewers. Signals in the port list should be easy to map back to the declarations, and the direction of each signal should be obvious before the body of the module is examined. A clear port definition is one of the simplest ways to keep a Verilog project maintainable as it grows.

4.5 Nets and Registers

Verilog separates physical connections from storage elements. Nets, such as `wire`, are used to model connections between components, while registers such as `reg` are used when the value must be held or assigned inside procedural code. This distinction is important because it reflects the hardware interpretation of the source. A net is driven by some source and used as a connection in the design, whereas a register is typically updated by an `always` or `initial` block during simulation.

Verilog also models four-valued logic, which is necessary when a design is not fully driven or when a signal cannot yet be resolved during simulation. The values are `0`, `1`, `x`, and `z`. The `z` state represents a high-impedance or undriven condition, while `x` represents an unknown or unresolved value. These states are useful when checking reset behavior, tri-state buses, or incomplete initialization. The following examples show the basic declaration style used in practice:

```

wire A, X, Y, Z, Cin, Cout;
wire [3:0] B, S;
wire [15:0] C;

reg q_out;
reg [7:0] counter;

```

Nets are not assigned inside procedural blocks, and registers are not used as direct continuous-connection targets in the same way as a wire. That separation keeps the model consistent with the intended hardware behavior and helps the simulator distinguish between continuous hardware connections and values that are updated under procedural control.

4.6 Numbers and Constants

Verilog numbers are written using a compact notation that encodes the size, radix, and value in a single literal. The general form is `<size>'<radix><value>`, where the radix indicates binary, octal, decimal, or hexadecimal representation. This syntax is used throughout RTL coding because it makes the intended width explicit and avoids ambiguity about sign extension or default integer size. The standard radix characters are `b`, `o`, `d`, and `h`.

Sized constants are preferable whenever the width matters, especially in module interfaces, state machines, and testbench stimulus. Verilog also allows unknown and high-impedance digits inside constants, which is useful when modeling incomplete bus values or uninitialized conditions during simulation. The examples below match the style used in the notes:

```
verilog
4'b1010
8'h3F
16'd255
4'b01xx
1'bz
```

Constants are not simply a syntactic detail. In a hardware description, they determine how the simulator interprets a value and how a synthesis tool treats a mask, width, or reset state. A value written without an explicit size may behave differently from the same value written with a fixed bit width, so laboratory code should state widths clearly whenever the design depends on them.

4.7 Operators

Verilog operators follow the same broad categories used in digital logic, but they are applied directly to hardware signals rather than to software variables. Bitwise operators act on individual bits, logical operators act on the truth value of an expression, shift operators move data left or right, and the conditional operator provides a compact form of multiplexing. Understanding this division is essential because a mistake in operator choice can change the synthesized hardware even when the simulation appears to run.

Bitwise operators operate on each bit of their operands. Unary reduction operators, by contrast, collapse a vector into a single result by combining all bits together. Logical operators return a true or false result and are typically used in conditions. The shift operators `<<` and `>>` move the bit pattern by a fixed number of positions, while the conditional operator `?:` is commonly used to express a multiplexer in a compact form. The basic usage is illustrated below:

```
verilog
assign A = X | (Y & ~Z);
assign B = &data_bus;
assign C = |mask_bus;
assign D = sel ? in1 : in0;

a = 4'b1010;
d = a >> 2;
c = a << 1;
```

In laboratory RTL, these operators are used constantly because they map directly onto the hardware view of the design. A conditional expression often describes the same logic that would be drawn as a multiplexer in a schematic, while bitwise and reduction operators are useful for masks, parity checks, and grouped control logic. The operator chosen in the source should therefore match the intended hardware structure, not merely the result visible in a quick simulation.

4.8 Laboratory Exercise – Half Adder

4.8.1 Objective

The purpose of this exercise is to design, simulate, and verify a Half Adder using Verilog HDL. The exercise introduces a complete but minimal digital design flow, beginning with the RTL description and continuing through testbench development, compilation, simulation, and waveform inspection in the Cadence environment.

A Half Adder is a suitable starting point because it is small enough to understand in a single session, yet complete enough to demonstrate the relationship between Boolean equations, Verilog code, stimulus generation, and simulation results. The same workflow used here will be used again in later chapters for larger combinational and sequential designs.

4.8.2 Theory

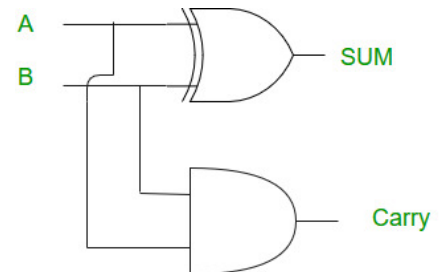
A Half Adder is a combinational circuit that adds two one-bit inputs, **A** and **B**, and produces two one-bit outputs: the *Sum* and the *Carry*. The Sum bit represents the least significant result of the addition, while the Carry bit indicates whether both inputs were high at the same time. Since the circuit has no storage element, its outputs depend only on the current input combination.

In Boolean form, the Half Adder is expressed as:

$$\text{Sum} = A \oplus B$$

$$\text{Carry} = A \cdot B$$

The Sum output is obtained using the exclusive-OR operation because the result is high only when exactly one input is high. The Carry output is obtained using the AND operation because a carry occurs only when both inputs are asserted simultaneously.



4.8.3 Truth Table

The truth table below lists all possible input combinations for a Half Adder and the corresponding Sum and Carry outputs. This table serves as the reference against which the Verilog model and the testbench results are checked.

A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Table 4.1: Truth table of the Half Adder.

4.8.4 Verilog Implementation

The Verilog description of a Half Adder is intentionally short, but it should still follow the same discipline used for larger RTL blocks. The module name identifies the design unit, the port list exposes the external interface, and the continuous assignments map directly to the Boolean expressions derived from the truth table. This makes the source easy to read and easy to verify against the expected hardware behavior.

```

module half_adder(A, B, Sum, Carry);
    input A, B;
    output Sum, Carry;

    assign Sum = A ^ B;
    assign Carry = A & B;
endmodule

```

verilog

The module declaration names the design and defines the port order. The input declarations identify the two one-bit operands. The output declarations identify the result signals that will be driven by the logic inside the module. The `assign` statement for `Sum` implements the exclusive-OR function, while the `assign` statement for `Carry` implements the AND function. The `endmodule` keyword closes the design unit and allows it to be instantiated inside a testbench or a larger circuit.

4.8.5 Testbench

A testbench is required because the design itself does not generate stimuli. The Half Adder must be exercised with all valid input combinations so that the simulation can confirm both outputs against the truth table. In a laboratory setting, the testbench also establishes a repeatable procedure for running the design inside the simulator, which is essential when the same flow must later be repeated for larger circuits.

```
verilog

`timescale 1ns/1ps

module half_adder_tb;
    reg A, B;
    wire Sum, Carry;

    half_adder dut (
        .A(A),
        .B(B),
        .Sum(Sum),
        .Carry(Carry)
    );

    initial begin
        $monitor($time, " A=%b B=%b Sum=%b Carry=%b", A, B, Sum, Carry);

        A = 0; B = 0; #10;
        A = 0; B = 1; #10;
        A = 1; B = 0; #10;
        A = 1; B = 1; #10;

        $finish;
    end
endmodule
```

The stimulus sequence applies each of the four possible input combinations in turn, with a short delay between each state so that the waveform viewer can capture the transitions clearly. The `$monitor` statement records the active values during simulation, making it easy to compare the observed output against the truth table. Once the final input combination has been applied, the simulation terminates with `$finish`.

4.8.6 Simulation Procedure

The simulation workflow should be treated as a standard laboratory procedure rather than as an ad hoc sequence of tool clicks. A clean project directory, a correct set of source files, and a known simulation entry point make it much easier to reproduce results later. The exact window layout may vary slightly between Cadence releases, but the workflow remains the same.

1. Create a new Cadence project or working directory for the exercise.
2. Add the design file containing the Half Adder RTL description.
3. Add the testbench file and ensure it references the design module correctly.
4. Compile the design and confirm that no syntax errors are reported.
5. Run the simulation and allow the testbench to apply all input combinations.
6. Open the waveform viewer and inspect the Sum and Carry signals.
7. Save the simulation session if the laboratory workflow requires a record of the results.

In practice, the compilation stage confirms that both files are readable by the simulator and that the module names, port names, and signal declarations agree. The simulation stage then checks that the testbench drives the design in the intended order. The waveform viewer is used at the end because it provides the clearest visual confirmation that the output transitions match the truth table.

4.8.7 Expected Results

The expected waveform is straightforward. When both inputs are low, both outputs remain low. When exactly one input is high, the Sum output becomes high while the Carry output remains low. When both inputs are high, the Sum output returns low and the Carry output becomes high. This is the defining behavior of a Half Adder and should be visible directly in the waveform viewer.

TODO: Insert Half Adder waveform screenshot

Figure 4.1: Expected Half Adder waveform showing Sum and Carry behavior for all input combinations.

The waveform should therefore show `Sum` following the XOR of `A` and `B`, while `Carry` follows the AND of the same inputs. If the observed outputs do not match the truth table, the design file and testbench should be checked first, followed by the simulation setup and signal connections.

4.8.8 Conclusion

This exercise demonstrates the complete Verilog verification flow on a simple combinational circuit. The Half Adder is small, but it introduces the essential sequence used throughout digital design: derive the logic, write the RTL, apply test stimulus, simulate the model, and verify the waveform against the expected behavior. That procedure will be reused in later chapters when the designs become larger and the simulation setup becomes more involved.

By completing the exercise, the reader gains a practical starting point for working with Cadence tools in a reproducible laboratory environment. The same method of checking the Boolean equations, validating the truth table, and confirming the waveform response will remain useful when the designs move from simple combinational logic to sequential circuits and larger RTL blocks.

05

Combinational Logic Design

System identity, licensing, and workstation preparation

5.1 Continuous Assignments

Combinational circuits produce outputs that depend only on the present values of their inputs. In Verilog, the most direct way to describe this kind of hardware is by using a continuous assignment. A continuous assignment represents a permanent connection from an expression to a net, similar to a wire driven by a gate or a small network of gates in a schematic.

The keyword used for this purpose is `assign`. It is written inside a module, but outside procedural blocks such as `initial` and `always`. The left-hand side of a continuous assignment is normally a net such as `wire`, and the right-hand side is an expression constructed from input signals, constants, and Verilog operators. Whenever any signal on the right-hand side changes, the expression is re-evaluated and the driven output is updated by the simulator. This continuous evaluation behavior matches the physical behavior of combinational logic, where gate outputs respond to changes at their inputs after propagation delay.

The simplest example is a two-input AND gate:

```
verilog
module and_gate(A, B, Y);
    input A, B;
    output Y;

    assign Y = A & B;
endmodule
```

The output `Y` is driven by the bitwise AND of `A` and `B`. If either input changes during simulation, `Y` is automatically updated. No explicit event control or procedural statement is required because the assignment is concurrent with the rest of the module.

An OR gate follows the same pattern:

```
verilog
module or_gate(A, B, Y);
    input A, B;
    output Y;

    assign Y = A | B;
endmodule
```

In this case, the operator `|` maps to a bitwise OR function. For one-bit signals it behaves as the OR gate used in a logic diagram. For vectors, the same operator applies bit by bit across corresponding positions of the operands.

Continuous assignments may also contain compound expressions. The following module drives three outputs from the same set of inputs:

```
verilog
module simple_logic(A, B, C, Y1, Y2, Y3);
  input A, B, C;
  output Y1, Y2, Y3;

  assign Y1 = A & B;
  assign Y2 = A | C;
  assign Y3 = (A & B) | (~C);
endmodule
```

Each `assign` statement describes a separate concurrent hardware connection. The expression for `Y3` contains an AND operation, an inversion, and an OR operation; a synthesis tool interprets this as the corresponding combinational network. The order of the statements in the source does not imply execution order. All continuous assignments are active at the same time, which is why they are commonly used for RTL descriptions of combinational logic.

5.2 Dataflow Modeling

Dataflow modeling describes a circuit by specifying how data values are transformed as they move from inputs to outputs. Instead of instantiating individual gates, the designer writes Boolean or arithmetic expressions that define the required output behavior. The synthesis tool then constructs the equivalent logic hardware from those expressions.

This is different from a structural description. In structural modeling, the source code explicitly connects gates or lower-level modules, much like a textual schematic. In dataflow modeling, the source code focuses on the relationship between signals. For example, a designer may write the expression for an output directly rather than listing every AND, OR, and NOT gate needed to realize it. This makes dataflow modeling concise and well suited to RTL design, especially for combinational blocks whose behavior is naturally expressed by equations.

A simple dataflow example is the implementation of an inverter and a two-input gate:

```
verilog
module basic_dataflow(A, B, Y_not, Y_and);
  input A, B;
  output Y_not, Y_and;

  assign Y_not = ~A;
  assign Y_and = A & B;
endmodule
```

The unary operator `~` describes bitwise inversion, while `&` describes bitwise AND. Since all signals are one bit wide in this example, each operator maps directly to the corresponding logic gate.

More useful combinational functions are usually written as expressions that combine several operators:

```

module combinational_network(A, B, C, D, F);
    input A, B, C, D;
    output F;

    assign F = (A & B) | (C & ~D);
endmodule

```

The output `F` becomes high when `A` and `B` are both high, or when `C` is high while `D` is low. Parentheses are used to make the intended grouping clear. Although Verilog has defined operator precedence rules, explicit grouping is preferred in laboratory code because it reduces ambiguity during review.

Dataflow descriptions also work naturally with buses. In the next example, the same operation is applied across four bits:

```

module vector_logic(A, B, mask, Y);
    input [3:0] A, B, mask;
    output [3:0] Y;

    assign Y = (A & mask) | (B & ~mask);
endmodule

```

Here, `A`, `B`, `mask`, and `Y` are four-bit vectors. The bitwise operators act independently on each bit position. This style is common in RTL because it describes a repeated logic pattern without writing a separate equation for every bit.

The following table summarizes the operators most frequently used in dataflow models of combinational logic.

Operator	Name	Typical Hardware Interpretation
<code>~</code>	Bitwise NOT	Inverter on each bit of the operand
<code>&</code>	Bitwise AND	AND gate for each bit position
<code> </code>	Bitwise OR	OR gate for each bit position
<code>^</code>	Bitwise XOR	XOR gate for each bit position
<code>+</code>	Addition	Combinational adder network
<code>?:</code>	Conditional	Multiplexer selection logic

Table 5.1: Common Verilog operators used in dataflow descriptions.

Dataflow modeling is widely used because it is compact, synthesizable, and close to the Boolean equations used during circuit design. Its main limitation is that very complex expressions can become difficult to read and debug. For larger designs, dataflow statements are often combined with hierarchy: small combinational functions are written as equations, and larger systems are built by instantiating those verified modules.

5.3 Conditional Operators

The conditional operator provides a compact way to select one of two expressions based on a condition. It is also called the ternary operator because it uses three operands: the condition, the expression selected when the condition is true, and the expression selected when the condition is false. Its general form is:

verilog

```
condition ? expression1 : expression2;
```

If `condition` evaluates to true, the result of the whole expression is `expression1`. If the condition evaluates to false, the result is `expression2`. In synthesizable combinational RTL, this operator is normally interpreted as selection logic. For one-bit selection, that hardware is a multiplexer.

A simple output selection example is shown below:

verilog

```
module conditional_example(A, B, sel, Y);  
    input A, B, sel;  
    output Y;  
  
    assign Y = sel ? A : B;  
endmodule
```

When `sel` is high, `Y` follows `A`. When `sel` is low, `Y` follows `B`. The statement is continuously evaluated, so any change on `sel`, `A`, or `B` is reflected at the output.

The selected expressions may also be logic expressions rather than simple signals:

verilog

```
module conditional_logic(A, B, C, mode, Y);  
    input A, B, C, mode;  
    output Y;  
  
    assign Y = mode ? (A & B) : (A | C);  
endmodule
```

In this example, `mode` selects between two different logic functions. When `mode` is high, the output is the AND of `A` and `B`. When `mode` is low, the output is the OR of `A` and `C`. This is useful when a circuit must route one of several possible results to a common output.

Conditional operators are also useful for signal routing on buses:

verilog

```
module bus_select(data_a, data_b, sel, data_out);  
    input [3:0] data_a, data_b;  
    input sel;  
    output [3:0] data_out;  
  
    assign data_out = sel ? data_a : data_b;  
endmodule
```

The selection is applied to the complete four-bit vector. Hardware synthesis produces four parallel one-bit selection circuits controlled by the same `sel` signal. This pattern is the natural Verilog form of a multiplexer, which is developed in detail in the next section.

5.4 Multiplexers

A multiplexer, commonly abbreviated as MUX, is a combinational circuit that selects one input from several available data inputs and forwards the selected value to a single output. The selection is controlled by one or more select lines. Multiplexers are fundamental building blocks in digital systems because they implement controlled routing: one data source is chosen while the others are ignored.

A multiplexer has three main parts: data inputs, select inputs, and an output. The number of select lines determines how many data inputs can be addressed. With one select line, two inputs can be selected. With two select lines, four inputs can be selected. In general, n select lines can address 2^n input choices. Since the output is determined only by the current values of the data and select inputs, a multiplexer is a combinational circuit and is well suited to continuous assignments and dataflow modeling.

5.4.1 2:1 Multiplexer

A 2:1 multiplexer has two data inputs, one select input, and one output. When the select input is low, the output follows I_0 . When the select input is high, the output follows I_1 . The complete truth table is shown in Table 5.2. The selected input determines the output value, while the unselected input has no effect on the output.

The Boolean expression for a 2:1 multiplexer is:

$$Y = \overline{sel} \cdot I_0 + sel \cdot I_1$$

The first product term enables I_0 when $sel = 0$, and the second product term enables I_1 when $sel = 1$.

sel	I0	I1	Y
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

Table 5.2: Full truth table of a 2:1 multiplexer.

The conditional operator gives the clearest Verilog description of this selection:

```

module mux2_conditional(I0, I1, sel, Y);
    input I0, I1, sel;
    output Y;

    assign Y = sel ? I1 : I0;
endmodule

```

verilog

The expression reads directly from the truth table. If sel is high, I_1 is selected; otherwise I_0 is selected. This is a dataflow description because the output is defined by an expression rather than by explicit gate instances.

The same 2:1 multiplexer can also be written using its Boolean equation:

verilog

```

module mux2_dataflow(I0, I1, sel, Y);
    input I0, I1, sel;
    output Y;

    assign Y = (~sel & I0) | (sel & I1);
endmodule

```

This version exposes the underlying AND-OR structure. The term `~sel & I0` enables `I0` when the select signal is low, and the term `sel & I1` enables `I1` when the select signal is high. The OR operation combines the enabled path into the output.

5.4.2 4:1 Multiplexer

A 4:1 multiplexer extends the same idea to four data inputs. It requires two select lines because four choices must be encoded. The select input is usually represented as a two-bit vector `sel[1:0]`, where `sel[1]` is the most significant select bit and `sel[0]` is the least significant select bit. The output follows exactly one of the four data inputs for each select combination.

The Boolean expression for a 4:1 multiplexer is:

$$Y = \overline{sel[1]} \cdot \overline{sel[0]} \cdot I0 + \overline{sel[1]} \cdot sel[0] \cdot I1 + sel[1] \cdot \overline{sel[0]} \cdot I2 + sel[1] \cdot sel[0] \cdot I3$$

Each product term corresponds to one decoded select condition. Only the selected data input is allowed to contribute to `Y`; the other data inputs are masked by the select logic.

sel[1]	sel[0]	I0	I1	I2	I3	Y
0	0	0	X	X	X	0
0	0	1	X	X	X	1
0	1	X	0	X	X	0
0	1	X	1	X	X	1
1	0	X	X	0	X	0
1	0	X	X	1	X	1
1	1	X	X	X	0	0
1	1	X	X	X	1	1

Table 5.3: Full truth table of a 4:1 multiplexer.

A compact dataflow implementation can be written by nesting conditional operators:

verilog

```

module mux4_conditional(I0, I1, I2, I3, sel, Y);
    input I0, I1, I2, I3;
    input [1:0] sel;
    output Y;

    assign Y = sel[1] ? (sel[0] ? I3 : I2) : (sel[0] ? I1 : I0);
endmodule

```

The outer conditional checks `sel[1]`. If `sel[1]` is low, the selection is between `I0` and `I1`; if `sel[1]` is high, the selection is between `I2` and `I3`. The inner conditional in each branch then uses `sel[0]` to choose the final input. This corresponds to a tree of 2:1 multiplexers.

The same function can be expressed using minterms:


```

verilog

module mux4_dataflow(I0, I1, I2, I3, sel, Y);
  input I0, I1, I2, I3;
  input [1:0] sel;
  output Y;

  assign Y = (~sel[1] & ~sel[0] & I0) |
             (~sel[1] & sel[0] & I1) |
             ( sel[1] & ~sel[0] & I2) |
             ( sel[1] & sel[0] & I3);

endmodule

```

This form is longer, but it makes the Boolean decoding of the select lines explicit. Each product term corresponds to one row of the truth table, and only one data input is enabled for each binary value of `sel`.

Multiplexers are used extensively in digital systems. They select operands for arithmetic units, choose between register outputs, route data onto internal buses, select next-state values in control logic, and implement configurable datapaths. The conditional operator and dataflow assignments introduced earlier provide the most direct Verilog notation for these functions.

5.5 Full Adder Design

A Half Adder adds two one-bit inputs and produces a Sum and Carry output. It is useful for introducing combinational design, but it cannot accept a carry from a lower-order bit position. Multi-bit addition requires that each bit position include the carry generated by the previous position. This is the purpose of a Full Adder.

A Full Adder has three one-bit inputs: `A`, `B`, and `Cin`. The input `Cin` represents the carry-in from a previous stage. It has two outputs: `Sum`, which is the one-bit addition result, and `Cout`, which is the carry-out passed to the next stage. The complete behavior is shown in Table 5.4.

A	B	Cin	Sum	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Table 5.4: Truth table of the Full Adder.

From the truth table, the Sum output is high when an odd number of inputs are high. The Carry output is high when at least two of the three inputs are high. The Boolean equations are:

$$\text{Sum} = A \oplus B \oplus \text{Cin}$$

$$\text{Cout} = (A \cdot B) + (B \cdot \text{Cin}) + (A \cdot \text{Cin})$$

The carry equation shows why the Full Adder is suitable for multi-bit arithmetic. A carry is produced either because `A` and `B` are both high, or because one operand bit is high while the incoming carry is also high. In a ripple carry adder, this `Cout` becomes the `Cin` of the next more significant Full Adder stage. This stage-to-stage dependency is called carry propagation.

A Full Adder can also be understood as two Half Adders and an OR gate. The first Half Adder adds `A` and `B`, producing an intermediate sum and carry. The second Half Adder adds the intermediate sum to `Cin`.

The two carry outputs are then ORed to produce Cout. In RTL, the same behavior can be written more directly using continuous assignments:

```
verilog

module full_adder(A, B, Cin, Sum, Cout);
    input A, B, Cin;
    output Sum, Cout;

    assign Sum = A ^ B ^ Cin;
    assign Cout = (A & B) | (B & Cin) | (A & Cin);
endmodule
```

The module declaration defines the Full Adder as a reusable design block. The input declaration lists the three one-bit operands, including the carry-in. The output declaration lists the two result signals. The first continuous assignment implements the three-input XOR function for Sum. The second continuous assignment implements the carry equation using AND terms for the possible carry-generating input pairs and an OR operation to combine them. Finally, endmodule closes the module so that it can be compiled independently or instantiated inside a larger design.

An alternative compact implementation uses Verilog addition and concatenation:

```
verilog

module full_adder_compact(A, B, Cin, Sum, Cout);
    input A, B, Cin;
    output Sum, Cout;

    assign {Cout, Sum} = A + B + Cin;
endmodule
```

The concatenation {Cout, Sum} forms a two-bit result. Since the maximum value of A + B + Cin is three, two bits are sufficient to represent both the carry and the sum. This form is concise and synthesizable, but the explicit Boolean version is often preferred when first studying the circuit because it shows the logic structure directly.

The Full Adder is the basic cell used to construct wider adders. In the following laboratory exercise, several Full Adder stages will be connected so that the carry-out of each stage feeds the carry-in of the next stage, forming a 4-bit ripple carry adder.

5.6 Laboratory Exercise – 4-Bit Ripple Carry Adder

5.6.1 Objective

The objective of this laboratory exercise is to design, simulate, and verify a 4-bit Ripple Carry Adder using Verilog HDL. The exercise combines the combinational design techniques introduced in this chapter and applies them to a practical arithmetic circuit.

By completing this exercise, the reader will demonstrate hierarchical design, module instantiation, carry propagation between adder stages, and functional verification through simulation. The design will be written as reusable Verilog modules and verified in the Cadence simulation environment using a separate testbench.

5.6.2 Theory

Binary addition is performed bit by bit, beginning with the least significant bit and progressing toward the most significant bit. At each bit position, the circuit must add the corresponding bits of the two operands and also include any carry generated by the previous lower-order position. A Half Adder is insufficient for this

task because it has no carry-in input.

A Full Adder solves this limitation by accepting three inputs: `A`, `B`, and `Cin`. It produces a `Sum` bit and a carry-out signal `Cout`. When several Full Adders are connected together, each stage handles one bit of a multi-bit addition. The carry-out of one stage becomes the carry-in of the next stage. This carry-chain structure is called a ripple carry adder because the carry information ripples from the least significant stage to the most significant stage.

The ripple carry adder is simple and well suited for laboratory study. It clearly shows module reuse, signal interconnection, and the relationship between a one-bit arithmetic cell and a wider arithmetic circuit.

5.6.3 Ripple Carry Adder Architecture

A 4-bit Ripple Carry Adder is constructed from four one-bit Full Adder modules connected in series. The first Full Adder adds `A[0]`, `B[0]`, and the external `Cin`. Its carry-out is connected to the carry-in of the second Full Adder. This pattern continues until the fourth stage produces the final carry-out `Cout`.

The input operands are `A[3:0]` and `B[3:0]`. The least significant bits are `A[0]` and `B[0]`, while the most significant bits are `A[3]` and `B[3]`. The output `Sum[3:0]` contains the four result bits, and `Cout` indicates whether the addition generated a carry beyond the most significant bit.

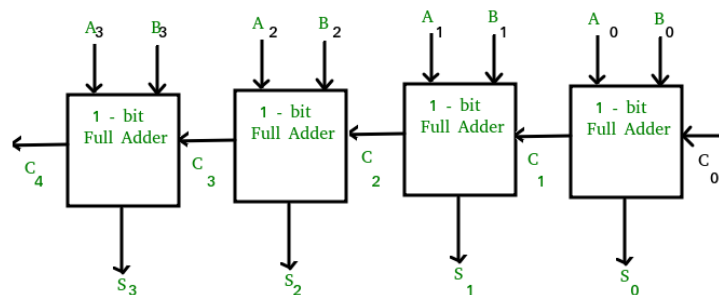


Figure 5.1: Conceptual block diagram of a 4-bit Ripple Carry Adder constructed from four Full Adder stages.

Data enters all four stages through the operand buses, but the carry path is sequential through the combinational chain. The first stage computes `Sum[0]` and an internal carry `c1`. The second stage uses `c1` to compute `Sum[1]` and `c2`. The third stage produces `Sum[2]` and `c3`. The fourth stage uses `c3` to compute `Sum[3]` and the final `Cout`. Although there is no clock in this circuit, the output cannot settle fully until the carry has propagated through the required stages.

5.6.4 Full Adder Module

The Full Adder module used in this exercise is the same one-bit combinational block discussed earlier in the chapter. It is written using continuous assignments so that the Boolean equations are represented directly in Verilog.

```

module full_adder(A, B, Cin, Sum, Cout);
    input A, B, Cin;
    output Sum, Cout;

    assign Sum = A ^ B ^ Cin;
    assign Cout = (A & B) | (B & Cin) | (A & Cin);
endmodule

```

verilog

The inputs `A` and `B` are the operand bits for the current stage, and `Cin` is the carry entering that stage. The output `Sum` is generated by the XOR of the three inputs. The output `Cout` is generated when any two or more inputs are high, as expressed by the three AND terms combined by OR operations. This module is designed as a reusable cell and will be instantiated four times in the 4-bit adder.

5.6.5 4-Bit Ripple Carry Adder Implementation

The top-level Ripple Carry Adder module demonstrates hierarchical combinational design. Instead of rewriting the Full Adder equations four times, the design instantiates four copies of the `full_adder` module and connects them through intermediate carry wires.

```

verilog

module ripple_carry_adder_4bit(A, B, Cin, Sum, Cout);
    input [3:0] A, B;
    input Cin;
    output [3:0] Sum;
    output Cout;

    wire c1, c2, c3;

    full_adder FA0 (
        .A(A[0]),
        .B(B[0]),
        .Cin(Cin),
        .Sum(Sum[0]),
        .Cout(c1)
    );

    full_adder FA1 (
        .A(A[1]),
        .B(B[1]),
        .Cin(c1),
        .Sum(Sum[1]),
        .Cout(c2)
    );

    full_adder FA2 (
        .A(A[2]),
        .B(B[2]),
        .Cin(c2),
        .Sum(Sum[2]),
        .Cout(c3)
    );

    full_adder FA3 (
        .A(A[3]),
        .B(B[3]),
        .Cin(c3),
        .Sum(Sum[3]),
        .Cout(Cout)
    );
endmodule

```

The module interface declares two four-bit input operands, one external carry-in, a four-bit sum output, and one final carry-out. The internal wires `c1`, `c2`, and `c3` carry information between adjacent Full Adder stages.

Instance `FA0` is the least significant stage. It adds `A[0]`, `B[0]`, and `Cin`, then produces `Sum[0]` and internal carry `c1`. Instance `FA1` uses `c1` as its carry-in and produces `c2`. Instance `FA2` uses `c2` and produces `c3`. Finally, instance `FA3` uses `c3` and produces the most significant sum bit and final

`Cout` . This carry chaining is the defining feature of the ripple carry architecture.

5.6.6 Testbench Development

A testbench is required because the Ripple Carry Adder module does not generate its own input stimulus. The testbench instantiates the design under test, applies representative operand values, waits for the combinational outputs to settle, and records the results for inspection in the simulator and waveform viewer.

The following testbench checks additions without carry generation, additions that produce internal carries, and additions that overflow beyond four bits.

```

verilog

`timescale 1ns/1ps

module ripple_carry_adder_4bit_tb;
    reg [3:0] A, B;
    reg Cin;
    wire [3:0] Sum;
    wire Cout;

    ripple_carry_adder_4bit dut (
        .A(A),
        .B(B),
        .Cin(Cin),
        .Sum(Sum),
        .Cout(Cout)
    );

    initial begin
        $monitor($time, " A=%b B=%b Cin=%b Cout=%b Sum=%b",
            A, B, Cin, Cout, Sum);

        A = 4'b0000; B = 4'b0000; Cin = 1'b0; #10;
        A = 4'b0011; B = 4'b0100; Cin = 1'b0; #10;
        A = 4'b0111; B = 4'b0001; Cin = 1'b0; #10;
        A = 4'b0101; B = 4'b0011; Cin = 1'b1; #10;
        A = 4'b1111; B = 4'b0001; Cin = 1'b0; #10;
        A = 4'b1111; B = 4'b1111; Cin = 1'b1; #10;

        $finish;
    end
endmodule

```

The first test applies zero operands and verifies the reset-like baseline behavior of the combinational circuit. The second test performs addition without overflow. The third and fourth tests require carry propagation through one or more stages. The fifth test produces a carry beyond the four-bit sum field, and the final test verifies the maximum input condition with an asserted carry-in. The `##10` delays separate each stimulus vector in simulation time so that the waveform viewer clearly shows the response to each input combination. The `$finish` system task terminates the simulation after all vectors have been applied.

5.6.7 Simulation Procedure

The simulation should be performed as a controlled laboratory workflow. Keep the Full Adder source, Ripple Carry Adder source, and testbench source in the same project directory or add all three files explicitly to the Cadence simulation setup.

1. Create a new Cadence project or working directory for the Ripple Carry Adder exercise.
2. Create the Verilog source file containing the `full_adder` module.
3. Create the Verilog source file containing the `ripple_carry_adder_4bit` module.

4. Create the testbench file containing `ripple_carry_adder_4bit_tb`.
5. Add all design and testbench files to the simulation environment.
6. Compile the source files and confirm that there are no syntax or elaboration errors.
7. Launch the simulation with `ripple_carry_adder_4bit_tb` as the top-level module.
8. Add `A`, `B`, `Cin`, `Sum`, and `Cout` to the waveform viewer.
9. Run the simulation long enough for all test vectors to execute.
10. Compare the waveform and console output against the expected binary addition results.

During compilation, errors are most commonly caused by mismatched module names, missing source files, or incorrect port names during instantiation. During simulation, incorrect results are usually caused by a broken carry connection between stages or a mismatch between the expected and actual bit ordering of the buses.

5.6.8 Expected Results

The simulation should show that `Sum[3:0]` contains the lower four bits of the binary addition and that `Cout` contains the carry beyond the most significant bit. For example, `0011 + 0100` should produce `Sum = 0111` and `Cout = 0`. The case `1111 + 0001` should produce `Sum = 0000` and `Cout = 1`, showing overflow beyond the four-bit output range.

Carry propagation should be visible in the functional behavior of the result. When a lower-order stage generates a carry, the next stage must include that carry in its own sum calculation. Although the internal carry wires may not be top-level ports, they can be added to the waveform viewer from the design hierarchy if the simulator setup allows internal signal probing.

TODO: Insert Ripple Carry Adder waveform

Figure 5.2: Expected waveform for the 4-bit Ripple Carry Adder showing operand inputs, carry-in, sum output, and carry-out.

The waveform verifies correct operation when each applied input vector produces the corresponding binary sum after the simulator advances past the stimulus delay. A mismatch in `Sum` indicates an error in the Full Adder equation or bit connection. A mismatch in `Cout` usually indicates an error in the final carry connection or in one of the intermediate carry wires.

5.6.9 Conclusion

This exercise demonstrated the design and verification of a 4-bit Ripple Carry Adder using Verilog HDL. The circuit was built hierarchically by defining a reusable Full Adder module and instantiating it four times inside a top-level combinational design. The intermediate carry wires illustrated how carry propagation links individual one-bit stages into a wider arithmetic circuit.

The testbench verified the design by applying representative input combinations and checking the resulting sum and carry-out behavior through simulation. This workflow reinforces the standard RTL design method used throughout digital systems engineering: describe a reusable module, connect modules hierarchically, compile the design, simulate with controlled stimulus, and verify the waveform against expected behavior. The same discipline will be used in later chapters when the manual moves from combinational circuits to sequential logic and more complex digital systems.

06

Sequential Logic Design

System identity, licensing, and workstation preparation

6.1 Why Memory is Required

The combinational circuits studied in the previous chapter produce outputs that depend only on the present values of their inputs. A multiplexer, Full Adder, or dataflow Boolean network has no internal record of what happened earlier. If the same inputs are applied again, the same outputs are produced, regardless of the previous input sequence. This property is useful for arithmetic and selection logic, but it is not sufficient for digital systems that must remember events, hold data, or move through a sequence of states.

Practical digital systems require *state*. A counter must remember its present count before it can generate the next count. A register must hold a binary value until the system is ready to load a new one. A finite state machine must remember its current state so that it can respond correctly to the next input condition. Digital controllers use this stored state to sequence operations, enable datapath blocks, and coordinate communication with other modules.

A circuit that includes memory is called a sequential circuit. Its output may depend on both the current inputs and the stored state. The stored state is normally updated at well-defined clock events so that the system changes in a controlled and predictable manner. This distinction is summarized in Table 6.1.

Feature	Combinational Logic	Sequential Logic
Output dependence	Present input values only	Present inputs and stored state
Memory	No internal memory	Contains storage elements
Typical examples	Adders, multiplexers, decoders	Registers, counters, finite state machines
Verilog style	Continuous assignments and combinational blocks	Clocked <code>always</code> blocks and non-blocking assignments

Table 6.1: Conceptual comparison between combinational and sequential logic.

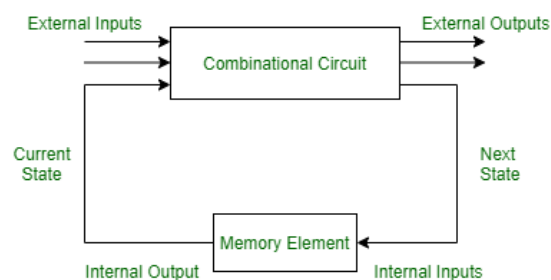


Figure 6.1: Conceptual of Sequential logic with stored state.

In RTL design, combinational logic and sequential logic are usually used together. Combinational logic computes next values, while storage elements capture those values at clock events. This chapter introduces the clocked storage elements required to build registers and counters, leading directly to the 4-bit synchronous counter exercise.

6.2 Clock Signals

A clock signal is a periodic digital signal used to coordinate when storage elements update their values. In a synchronous digital system, flip-flops and registers usually sample their inputs on a specific clock transition. This shared timing reference allows many storage elements to update together and makes the behavior of the system easier to simulate, verify, and synthesize.

The *clock period* is the time required for one complete clock cycle. The *clock frequency* is the number of cycles per second and is the reciprocal of the period. A clock transition from low to high is called a rising edge, while a transition from high to low is called a falling edge. In Verilog RTL, these transitions are written using `posedge clk` and `negedge clk`.

TODO: Insert clock waveform diagram

Figure 6.2: Clock waveform showing period, frequency, rising edge, and falling edge.

A clock can be generated in a testbench using an `always` block and a delay. The following example toggles `clk` every 5 ns, producing a clock period of 10 ns.

```
verilog
`timescale 1ns/1ps

module clock_example_tb;
    reg clk;

    initial begin
        clk = 1'b0;
    end

    always #5 clk = ~clk;
endmodule
```

This form is intended for simulation testbenches, not for synthesizable design logic. It creates a simple periodic signal that can drive clocked modules during verification.

A storage element responds to a clock edge by listing that edge in the sensitivity list of an `always` block:

```
verilog

always @(posedge clk) begin
    q <= d;
end
```

The statement above means that `q` is updated only on the rising edge of `clk`. If a falling-edge triggered design is required, `negedge clk` is used instead. Synchronous systems are preferred in most RTL designs because the state updates occur at known clock boundaries, which simplifies simulation, timing analysis, and debugging.

6.3 D Flip-Flops

A D Flip-Flop is the basic edge-triggered storage element used in synchronous RTL design. It has a data input, usually named `D`, a clock input, and an output, usually named `Q`. On the active clock edge, the value present at `D` is captured and stored at `Q`. Between active clock edges, `Q` holds its previous value even if `D` changes.

This behavior is what gives sequential circuits memory. The flip-flop does not continuously pass its input to the output. Instead, it samples the input at a clock transition and holds that sampled value until the next active edge. Table 6.2 summarizes the functional behavior for a rising-edge triggered D Flip-Flop.

Clock Event	D	Next Q
Rising edge	0	0
Rising edge	1	1
No rising edge	X	Previous Q

Table 6.2: Functional behavior of a rising-edge triggered D Flip-Flop.

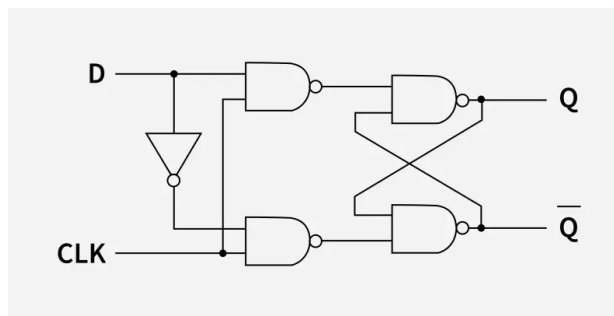


Figure 6.3: D Flip Flop Circuit Diagram

The timing behavior should be interpreted carefully in simulation. The input `D` may change many times between clock edges, but only the value present at the active edge is stored. After the edge, `Q` remains stable until the next active edge, except when a reset signal is included and asserted.

```

module d_flip_flop(d, clk, q);
    input d, clk;
    output reg q;

    always @(posedge clk) begin
        q <= d;
    end
endmodule

```

verilog

The `always` block describes logic that is evaluated when the event in its sensitivity list occurs. In this example, the event is the rising edge of `clk`. The output `q` is declared as `reg` because it is assigned inside a procedural block. The assignment operator `<=` is a non-blocking assignment and is the standard form used for clocked sequential logic. It allows all registers triggered by the same clock edge to be updated together in simulation, matching the behavior expected from synchronous hardware.

Most practical flip-flops include a reset signal so that the circuit can be placed into a known state. A synchronous reset is checked inside the clocked block and affects the output only on the active clock edge.

verilog

```

module dff_sync_reset(d, clk, reset, q);
    input d, clk, reset;
    output reg q;

    always @(posedge clk) begin
        if (reset)
            q <= 1'b0;
        else
            q <= d;
    end
endmodule

```

In this module, `reset` is sampled on the rising edge of `clk`. If `reset` is high at that edge, `q` is cleared to zero. Otherwise, `q` captures `d`.

An asynchronous reset is included in the sensitivity list and can clear the output without waiting for the next clock edge. This is common when a design must enter a known state immediately after reset is asserted.

verilog

```

module dff_async_reset(d, clk, resetn, q);
    input d, clk, resetn;
    output reg q;

    always @(posedge clk or negedge resetn) begin
        if (!resetn)
            q <= 1'b0;
        else
            q <= d;
    end
endmodule

```

The signal `resetn` is active low, as indicated by the suffix `n` and by the condition `!resetn`. The block executes either on a rising clock edge or on a falling edge of reset. If reset is asserted, the output clears immediately in simulation. If reset is not asserted, the flip-flop behaves like the basic clocked D Flip-Flop.

6.4 Registers

A register is a collection of flip-flops used to store a multi-bit binary value. Each bit of the register is held by one flip-flop, and all bits are normally loaded on the same active clock edge. For example, a 4-bit register stores four bits, an 8-bit register stores eight bits, and an `n`-bit register stores `n` bits.

Registers are used throughout digital systems. They hold operands in datapaths, store intermediate results, capture status flags, preserve configuration values, and form the state registers of finite state machines. In a synchronous RTL design, registers define the boundaries between cycles: combinational logic computes a value during one cycle, and a register captures that value at the next active clock edge.

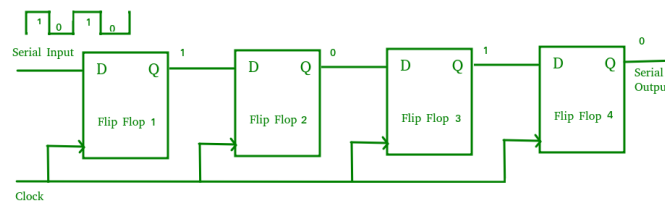


Figure 6.4: Register architecture showing multiple D Flip-Flops sharing a common clock and reset.

In Verilog, a register output assigned inside a clocked `always` block is declared using `reg`. The width of the stored value is written using a vector range.

```
verilog
reg [3:0] data_reg;
reg [7:0] status_reg;
```

A simple 4-bit register with synchronous reset and clocked loading is shown below.

```
verilog
module register_4bit(d, clk, reset, q);
    input [3:0] d;
    input clk, reset;
    output reg [3:0] q;

    always @(posedge clk) begin
        if (reset)
            q <= 4'b0000;
        else
            q <= d;
    end
endmodule
```

On each rising edge of `clk`, the register either clears to zero or captures the four-bit input `d`. The output `q` then holds that value until the next clock edge. The same pattern can be extended to wider registers by changing the vector width.

```
verilog
module register_8bit(d, clk, resetn, q);
    input [7:0] d;
    input clk, resetn;
    output reg [7:0] q;

    always @(posedge clk or negedge resetn) begin
        if (!resetn)
            q <= 8'b00000000;
        else
            q <= d;
    end
endmodule
```

This 8-bit register uses an active-low asynchronous reset. The design style is the same as the D Flip-Flop example, but the stored signal is now a vector. In datapath design, such registers are commonly placed before and after combinational blocks so that values move through the system one clock cycle at a time.

6.5 Counters

A counter is a sequential circuit that moves through a defined sequence of binary states. The most common counter increments by one on each active clock edge, but counters may also decrement, load a specified value, enable or disable counting, or count through a custom sequence. Since a counter must remember its current count before computing the next count, it is built from registers and combinational next-state logic.

Counters are used for timing generation, event counting, address sequencing, loop control, clock division, and control-state tracking. A binary up-counter with `n` bits can represent `2n` states. When the counter reaches its maximum value and increments again, it wraps around to zero unless additional logic is provided.

Asynchronous Counters

In an asynchronous counter, also called a ripple counter, only the first flip-flop is driven by the external clock. The output of one flip-flop clocks the next stage. This creates a ripple effect: later bits change only after earlier bits have toggled. The circuit is simple, but the accumulated propagation delay makes timing less predictable, especially as the counter width increases. For this reason, asynchronous counters are generally avoided in synthesizable RTL intended for modern synchronous digital systems.

Synchronous Counters

In a synchronous counter, all flip-flops share the same clock. The next count value is computed by combinational logic and captured by all counter bits on the same active clock edge. This makes the timing behavior easier to analyze and verify. Synchronous counters are therefore the preferred style for RTL design and are the style used in the following laboratory exercise.

Table 6.3 shows the state sequence for a simple 3-bit binary up-counter.

Present Count	Next Count
000	001
001	010
010	011
011	100
100	101
101	110
110	111
111	000

Table 6.3: State transition sequence of a 3-bit binary up-counter.

A basic synchronous counter can be written by registering the count value and adding one on each clock edge. The increment operation is a combinational addition whose result is captured by the register at the active edge.

```

verilog

module counter_3bit(clk, reset, count);
    input clk, reset;
    output reg [2:0] count;

    always @(posedge clk) begin
        if (reset)
            count <= 3'b000;
        else
            count <= count + 1'b1;
        end
    endmodule

```

When `reset` is high at the rising edge of `clk`, the counter returns to zero. Otherwise, the current value of `count` is incremented by one. Since `count` is three bits wide, the value after `111` wraps around to `000`. This wrap-around behavior is a natural result of fixed-width binary arithmetic in Verilog.

The next laboratory exercise applies these ideas to a complete 4-bit synchronous counter. The design will use a clocked register, reset behavior, and binary increment logic, then verify the counting sequence using simulation waveforms in the Cadence environment.

6.6 Laboratory Exercise – 4-Bit Synchronous Counter

6.6.1 Objective

The objective of this experiment is to design, simulate, and verify a 4-bit synchronous counter using Verilog HDL. The exercise applies the sequential logic concepts introduced in this chapter and demonstrates how a clocked RTL block stores and updates state over time.

By completing this exercise, the reader will implement a state-based digital system, observe clocked operation, use sequential logic design practices, and verify the counter behavior through simulation. The completed design will be suitable for compilation and waveform inspection in the Cadence simulation environment.

6.6.2 Theory

A binary counter advances through a sequence of binary states. For a 4-bit up-counter, the output can represent sixteen values, from `0000` to `1111`. On each active clock edge, the counter increments by one and stores the new value in its internal register. Since the counter must remember its present value before producing the next value, it is a sequential circuit rather than a purely combinational circuit.

In a synchronous counter, all flip-flops share the same clock signal. This means that all bits of the counter update together on the active clock edge. The count sequence progresses as:

```
terminal
0000
0001
0010
0011
...
1111
0000
```

After the maximum 4-bit value `1111`, the next increment produces `0000`. This is called overflow or wrap-around behavior. It occurs because the counter register has a fixed width of four bits; any carry beyond the most significant bit is discarded unless it is explicitly captured by additional logic.

6.6.3 Counter Architecture

The 4-bit synchronous counter contains a clock input, a reset input, and a 4-bit output bus. The output bus represents the present count value. Internally, the counter behaves as a 4-bit register whose next value is either zero during reset or the present count plus one during normal operation.

All storage bits receive the same clock. On each rising edge of the clock, the counter checks the reset signal. If reset is asserted, the count is cleared to `0000`. If reset is not asserted, the counter advances to the next binary value.

TODO: Insert 4-Bit Synchronous Counter block diagram

Figure 6.5: Block diagram of a 4-bit synchronous counter with clock, reset, and count output.

Table 6.4 shows several successive states of the 4-bit counter.

Present Count	Next Count
0000	0001
0001	0010
0010	0011
0011	0100
0100	0101
1110	1111
1111	0000

Table 6.4: Selected state transitions of a 4-bit synchronous up-counter.

6.6.4 Verilog Implementation

The RTL implementation uses one clocked `always` block. The counter output is declared as a 4-bit `reg` because it is assigned inside the procedural block. Non-blocking assignments are used so that the update behavior matches edge-triggered hardware.

```

verilog
module synchronous_counter_4bit(clk, reset, count);
    input clk, reset;
    output reg [3:0] count;

    always @(posedge clk) begin
        if (reset)
            count <= 4'b0000;
        else
            count <= count + 1'b1;
    end
endmodule

```

The module interface contains the clock input `clk`, the reset input `reset`, and the 4-bit output `count`. The `always` block executes on each rising edge of `clk`. If `reset` is high at that edge, the counter is loaded with `0000`. Otherwise, the present value of `count` is incremented by one. Since `count` is four bits wide, the increment after `1111` wraps around to `0000`.

6.6.5 Testbench Development

Verification is required to confirm that the counter initializes correctly, increments on clock edges, and responds properly to reset. The testbench generates a clock, applies reset, lets the counter run for several cycles, applies reset again, and then terminates the simulation.

```

verilog
`timescale 1ns/1ps

module synchronous_counter_4bit_tb;
    reg clk, reset;
    wire [3:0] count;

    synchronous_counter_4bit dut (
        .clk(clk),
        .reset(reset),
        .count(count)
    );

    initial begin

```

```

    clk = 1'b0;
end

always #5 clk = ~clk;

initial begin
    $monitor($time, " reset=%b count=%b", reset, count);

    reset = 1'b1;
    #12;

    reset = 1'b0;
    #180;

    reset = 1'b1;
    #10;

    reset = 1'b0;
    #40;

    $finish;
end
endmodule

```

The `always #5` statement generates a 10 ns clock period for simulation. The first reset interval initializes the counter to zero. After reset is released, the counter advances on each rising clock edge. The second reset interval verifies that the counter can be returned to `0000` after it has already been counting. The `$monitor` statement prints the reset and count values during simulation, while the waveform viewer provides the clearest visual confirmation of the state transitions.

6.6.6 Simulation Procedure

Use the following workflow to perform the experiment in the Cadence simulation environment:

1. Create a new working directory or Cadence project for the synchronous counter experiment.
2. Create the Verilog source file containing `synchronous_counter_4bit`.
3. Create the testbench file containing `synchronous_counter_4bit_tb`.
4. Add both Verilog files to the simulation setup.
5. Compile the design and confirm that no syntax or elaboration errors are reported.
6. Run the simulation with `synchronous_counter_4bit_tb` as the top-level module.
7. Open the waveform viewer and add `clk`, `reset`, and `count`.
8. Verify that the counter resets to zero and increments on rising clock edges.

If the design does not compile, check the module names, port names, and file order. If the simulation runs but the waveform is incorrect, inspect the reset polarity, the clock generation statement, and the use of non-blocking assignments in the counter.

6.6.7 Expected Results

During simulation, the counter output should become `0000` whenever reset is asserted and a rising clock edge occurs. After reset is released, the output should increment by one on each rising edge of `clk`. The sequence should progress through the normal binary count values from `0000` to `1111`. After `1111`, the next increment should wrap around to `0000`.

The waveform should show that changes in `count` occur only at rising clock edges. This confirms that the design is sequential and clock controlled, rather than continuously driven like combinational logic.

TODO: Insert synchronous counter waveform

Figure 6.6: Expected waveform of the 4-bit synchronous counter showing reset behavior, binary incrementing, and wrap-around.

Correct operation is verified when the waveform shows initialization to `0000`, steady binary progression at clock edges, reset returning the count to zero, and overflow from `1111` back to `0000`.

6.6.8 Conclusion

This experiment demonstrated the design and verification of a 4-bit synchronous counter using Verilog HDL. The counter was implemented as a clocked sequential circuit using an `always @(posedge clk)` block, non-blocking assignments, reset logic, and fixed-width binary incrementing.

The simulation verified clocked state transitions, reset behavior, normal counter operation, and wrap-around after the maximum 4-bit value. These concepts form the basis for more advanced sequential systems, including finite state machines, digital controllers, and processing datapaths that will be explored in later chapters.

Part III

VERIFICATION

Testbenches, simulation, waveform analysis, and coverage

07

Verilog Testbenches

System identity, licensing, and workstation preparation

7.1 What is Verification?

7.2 Testbench Architecture

7.3 Initial Blocks

7.4 Always Blocks

7.5 Clock Generation

7.6 Stimulus Generation

7.7 Self-Checking Testbenches

7.8 Laboratory Exercise

Counter Testbench

08

Simulation and Waveform Analysis

System identity, licensing, and workstation preparation

8.1 Compilation

8.2 Elaboration

8.3 Simulation

8.4 Understanding Waveforms

8.5 Common RTL Bugs

8.6 Debugging Methodology

8.7 Laboratory Exercise

Counter Waveform Analysis

09

Coverage Analysis using IMC

System identity, licensing, and workstation preparation

9.1 Why Simulation Alone is Not Enough

9.2 Code Coverage

9.3 Branch Coverage

9.4 Toggle Coverage

9.5 Functional Coverage

9.6 Using IMC

9.7 Coverage Reports

9.8 Coverage Closure

9.9 Laboratory Exercise

Counter Coverage Analysis

Part IV

LOGIC SYNTHESIS

Synthesis, timing analysis, and logical equivalence checking

10

Introduction to Synthesis

System identity, licensing, and workstation preparation

10.1 RTL vs Hardware

10.2 Standard Cell Libraries

10.3 Liberty Files

10.4 Timing Constraints

10.5 The Synthesis Flow

10.6 Reading Reports

11

Logic Synthesis using Genus

System identity, licensing, and workstation preparation

11.1 Preparing RTL

11.2 Writing Constraints

11.3 Running Genus

11.4 Generated Outputs

- Netlist
- Timing Reports
- Area Reports
- Power Reports

11.5 Laboratory Exercise

Synthesizing the Counter

12

Timing Analysis

System identity, licensing, and workstation preparation

12.1 Propagation Delay

12.2 Setup Time

12.3 Hold Time

12.4 Clock Frequency

12.5 Critical Paths

12.6 Slack

12.7 Laboratory Exercise

Counter Timing Analysis

13

Logical Equivalence Checking

System identity, licensing, and workstation preparation

13.1 Why Equivalence Checking Matters

13.2 Golden vs Revised Designs

13.3 Conformal Workflow

13.4 Laboratory Exercise

Verifying Counter Equivalence

Part V

PHYSICAL DESIGN

Floorplanning, power planning, placement, clocking, routing, and verification

14

Introduction to Physical Design

System identity, licensing, and workstation preparation

14.1 What Happens After Synthesis?

14.2 Netlists and Standard Cells

14.3 Die, Core and IO Regions

14.4 Physical Design Flow Overview

15

Floorplanning

System identity, licensing, and workstation preparation

15.1 Defining the Core Area

15.2 IO Placement

15.3 Macro Placement

15.4 Utilization

15.5 Laboratory Exercise

Counter Floorplan

16

Power Planning

System identity, licensing, and workstation preparation

16.1 Why Chips Need Power Networks

16.2 Power Rings

16.3 Power Stripes

16.4 IR Drop Concepts

16.5 Ground Networks

16.6 Laboratory Exercise

Counter Power Plan

17

Placement

System identity, licensing, and workstation preparation

17.1 Standard Cell Placement

17.2 Congestion

17.3 Density Optimization

17.4 Laboratory Exercise

Counter Placement

18

Clock Tree Synthesis

System identity, licensing, and workstation preparation

18.1 Clock Distribution

18.2 Clock Buffers

18.3 Clock Skew

18.4 Clock Latency

18.5 Laboratory Exercise

Counter CTS

19

Routing

System identity, licensing, and workstation preparation

19.1 Global Routing

19.2 Detailed Routing

19.3 Routing Resources

19.4 Signal Integrity

19.5 Laboratory Exercise

Counter Routing

20

Physical Verification

System identity, licensing, and workstation preparation

20.1 Design Rule Checking (DRC)

20.2 Layout Versus Schematic (LVS)

20.3 Common Physical Violations

20.4 Laboratory Exercise

Counter Verification

Part VI

TAPEOUT AND BEYOND

GDSII generation, packaging context, and complete flow review

21

GDSII Generation

System identity, licensing, and workstation preparation

21.1 What is GDSII?

21.2 Manufacturing Data

21.3 Tapeout Process

22

From Silicon to PCB

System identity, licensing, and workstation preparation

22.1 Packaging

22.2 Pin Assignment

22.3 Chip Integration

22.4 PCB Design Workflow

22.5 Why PCB Design is Different from ASIC Design

23

Complete Design Flow Review

System identity, licensing, and workstation preparation

23.1 Revisiting the Counter

23.2 RTL → Simulation → Coverage

23.3 Synthesis → Timing → Verification

23.4 Floorplanning → CTS → Routing

23.5 Tapeout

23.6 Final Flow Summary

Verilog → Simulation → IMC → Genus → Conformal → Innovus → GDSII

01

c2ssetup Script for Automating User/Host-name Config

System identity, licensing, and workstation preparation

```
#!/bin/csh

echo "====="
echo " Available Login Users"
echo "====="

awk -F: '$3 >= 1000 && $3 < 65534 && $1 != "nobody" {print $1}' /etc/passwd

echo ""
echo -n "Enter username to rename: "
set OLD_USER = "$<"

id $OLD_USER >& /dev/null
if ( $status != 0 ) then
    echo "Error: User '$OLD_USER' does not exist."
    exit 1
endif

echo ""
echo -n "Enter new username: "
set NEW_USER = "$<"

echo ""
echo -n "Enter display name (Full Name): "
set DISPLAY_NAME = "$<"

echo ""
echo -n "Enter new hostname: "
set NEW_HOST = "$<"

set OLD_GROUP = `id -gn $OLD_USER`

echo ""
echo "====="
echo " Renaming User"
echo "====="

sudo usermod -l $NEW_USER $OLD_USER
```

```

if ( $status != 0 ) then
    echo "Failed to rename user."
    exit 1
endif

echo ""
echo "=====
echo " Updating Home Directory"
echo "=====

sudo usermod -d /home/$NEW_USER -m $NEW_USER

if ( $status != 0 ) then
    echo "Failed to update home directory."
    exit 1
endif

echo ""
echo "=====
echo " Renaming Primary Group"
echo "=====

sudo groupmod -n $NEW_USER $OLD_GROUP

if ( $status != 0 ) then
    echo "Failed to rename group."
    exit 1
endif

echo ""
echo "=====
echo " Updating Display Name"
echo "=====

sudo usermod -c "$DISPLAY_NAME" $NEW_USER

if ( $status != 0 ) then
    echo "Failed to update display name."
    exit 1
endif

echo ""
echo "=====
echo " Fixing Ownership"
echo "=====

sudo chown -R ${NEW_USER}:${NEW_USER} /home/$NEW_USER

if ( -d /home/install ) then
    sudo chown -R ${NEW_USER}:${NEW_USER} /home/install
endif

echo ""
echo "=====
echo " Updating Hostname"
echo "=====

echo "$NEW_HOST" | sudo tee /etc/hostname > /dev/null

sudo sed -i "s/^127\.0\.0\.1.*$/127.0.0.1 localhost loghost $NEW_HOST/" /etc/hosts

grep -q "c2s.cdacb.in" /etc/hosts
if ( $status != 0 ) then

```

```

    echo "14.139.1.126 c2s.cdacb.in" | sudo tee -a /etc/hosts > /dev/null
endif

grep -q "cadence_server" /etc/hosts
if ( $status != 0 ) then
    echo "14.139.1.126 cadence_server" | sudo tee -a /etc/hosts > /dev/null
endif

sudo hostnamectl set-hostname "$NEW_HOST"

if ( $status != 0 ) then
    echo "Failed to set hostname."
    exit 1
endif

echo ""
echo "=====
echo " Setup Complete"
echo "=====

echo "Username      : $NEW_USER"
echo "Display Name   : $DISPLAY_NAME"
echo "Hostname       : $NEW_HOST"

echo ""
echo "Please log out and log back in."
echo ""

# For Synopsys tools add to ~/.cshrc if required:
#
# setenv SNPSLMD_LICENSE_FILE 27020@14.139.1.126
# setenv FLEXLM_DIAGNOSTICS 3

```

Save this file as `c2ssetup.sh` and then run the following command to make it executable:

```
sudo chmod +x c2ssetup.sh
```

Run the file:

```
./c2ssetup.sh
```


02

Cshrc configuration code

System identity, licensing, and workstation preparation